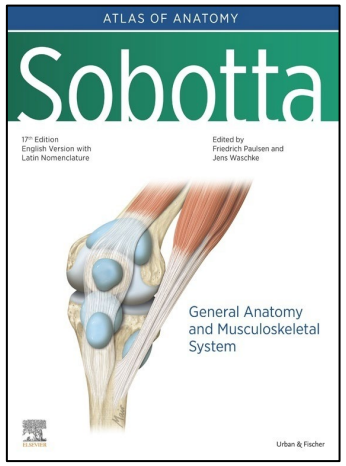
Modul 5.3 Bevægeapparatet

Columnas (COL) - systematiske anatomi





Tong Tong
Assistant Professor

Normal text = Danish
Normal text + Italic = Latin anatomical term
Italic + smaller + green = English

Læringsmål – Systematisk anatomi

Learning Objectives – Systematic Anatomy

- **Beskrive rygsøjlens regioner, hvirvler og fysiologiske krumninger.**

  Describe spinal regions, vertebral morphology, and physiological curves.

- **Forklare facetled, ligamenter og disci samt deres funktion.**

  Explain facet joints, ligaments, discs and their functions.

- **Identificere overfladiske og dybe rygmuskler og deres hovedfunktion.**

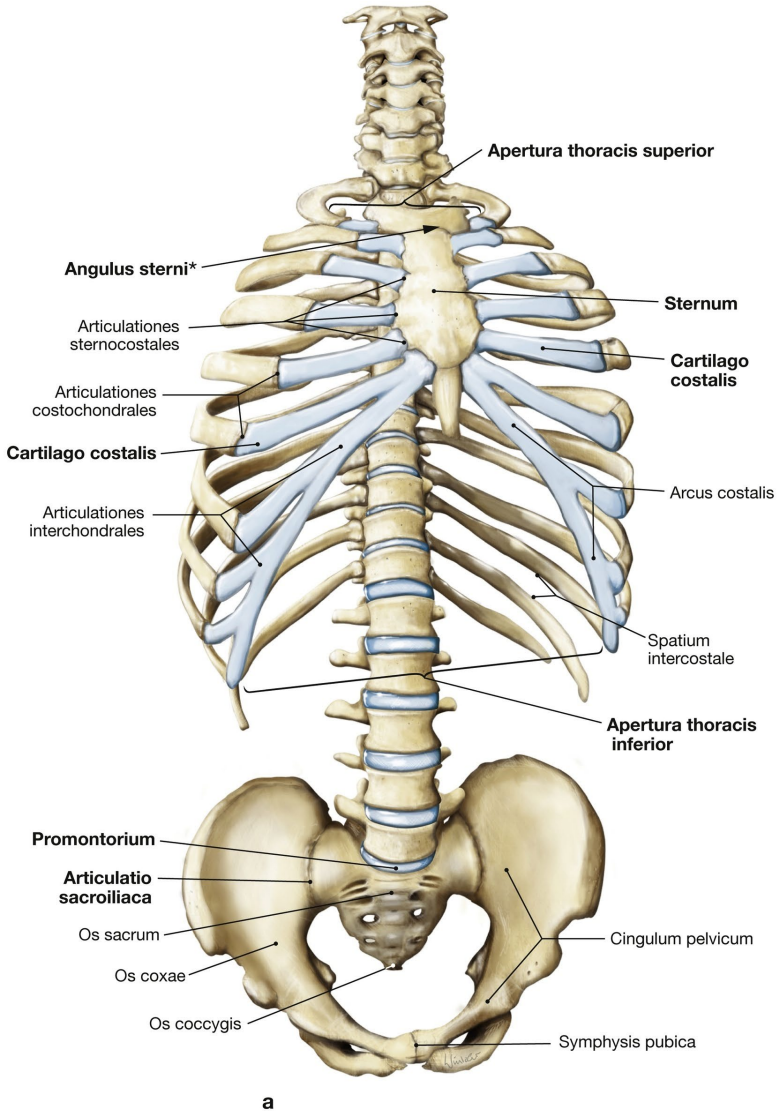
  Identify major extrinsic/intrinsic back muscles and their actions.

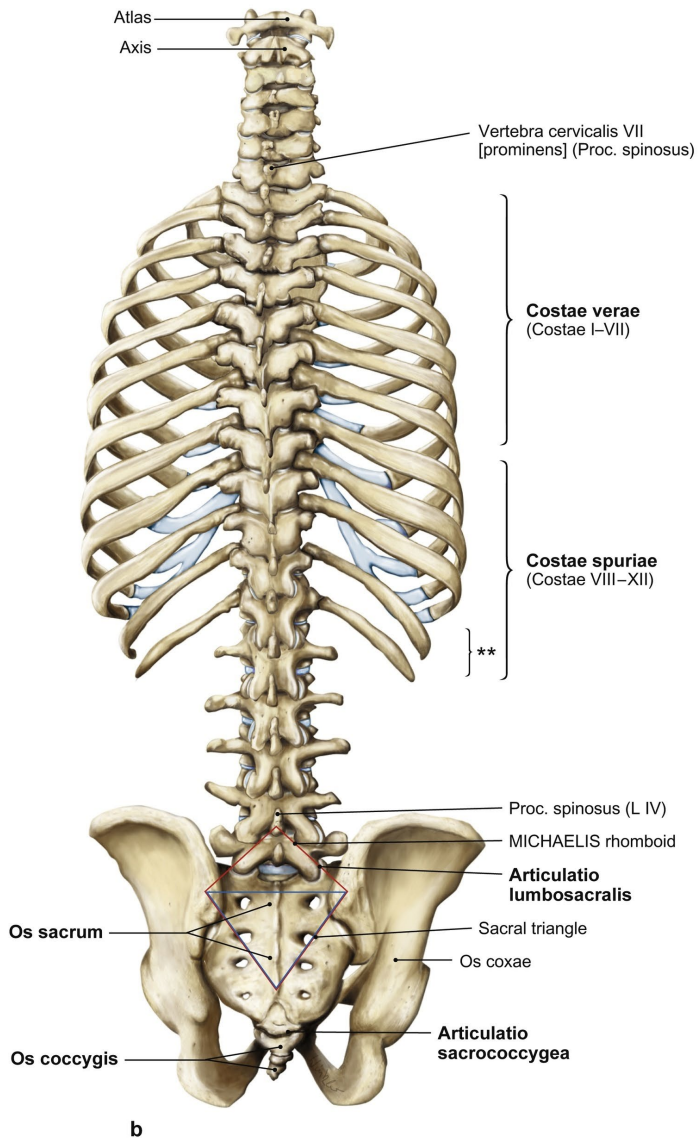
- **Forklare fleksion, ekstension, lateralfleksion og rotation.**

  Explain spinal flexion, extension, lateral flexion and rotation.

Knogler og brusk i skelettet i truncus

Bone and cartilage of the skeleton of the trunk





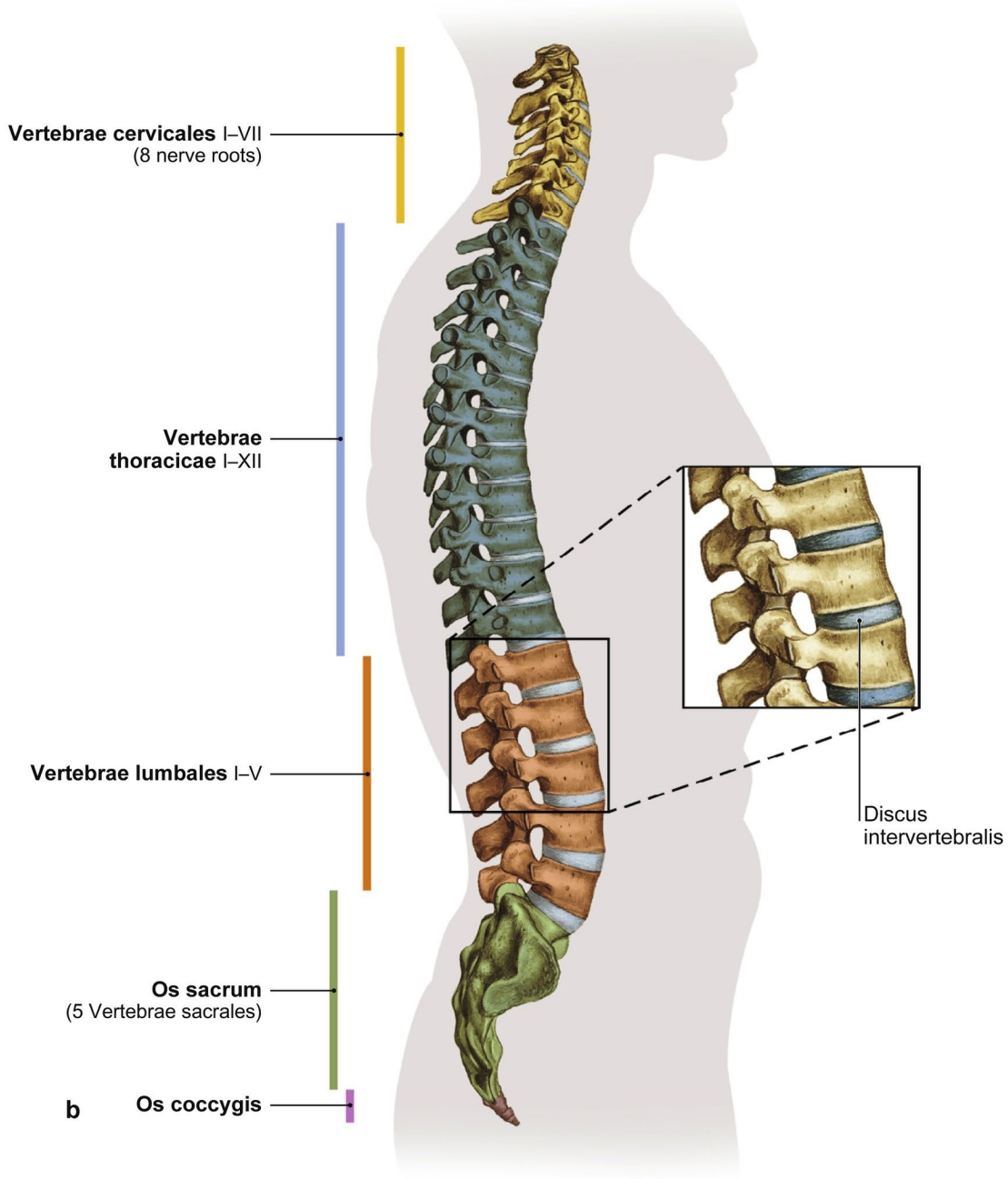
Columna vertebralis – Overblik
Vertebral column – Overview

- Støtte: Bærer kroppens vægt.
  Support: Bears body weight.

- Bevægelighed: Hvirvler (vertebrae), disci og rygmuskler muliggør bevægelse.
  Mobility: Vertebrae, discs, and back muscles enable movement.

- Beskyttelse: Canalis vertebralis beskytter rygmarven (medulla spinalis).
  Protection: The vertebral canal protects the spinal cord.

- Belastningsfordeling: Disci intervertebrales og rygsøjlens krumninger fordeler belastningen.
  Load distribution: Intervertebral discs and spinal curvatures distribute loads.

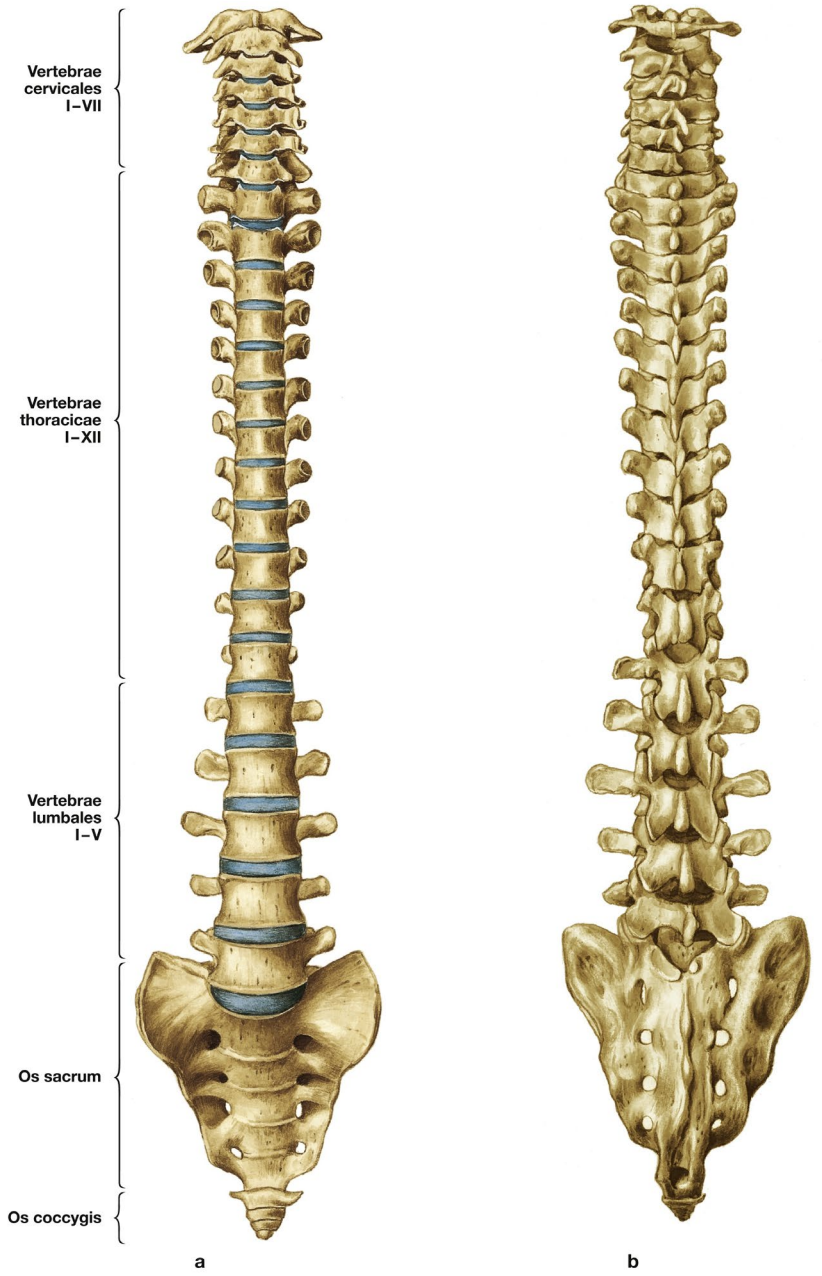


Columna vertebralis – Opbygning
Vertebral Column – Structure

- 24 frie hvirvler (vertebrae presacrales)
  - 7 halshvirvler (vertebrae cervicales)
    7 cervical vertebrae
  - 12 brysthvirvler (vertebrae thoracicae)
    12 thoracic vertebrae
  - 5 lændehvirvler (vertebrae lumbales)
    5 lumbar vertebrae
- 2 sammenvoksede knogleafsnit
  - Korsben (os sacrum)
    Sacrum
  - Haleben (os coccygis)
    Coccyx

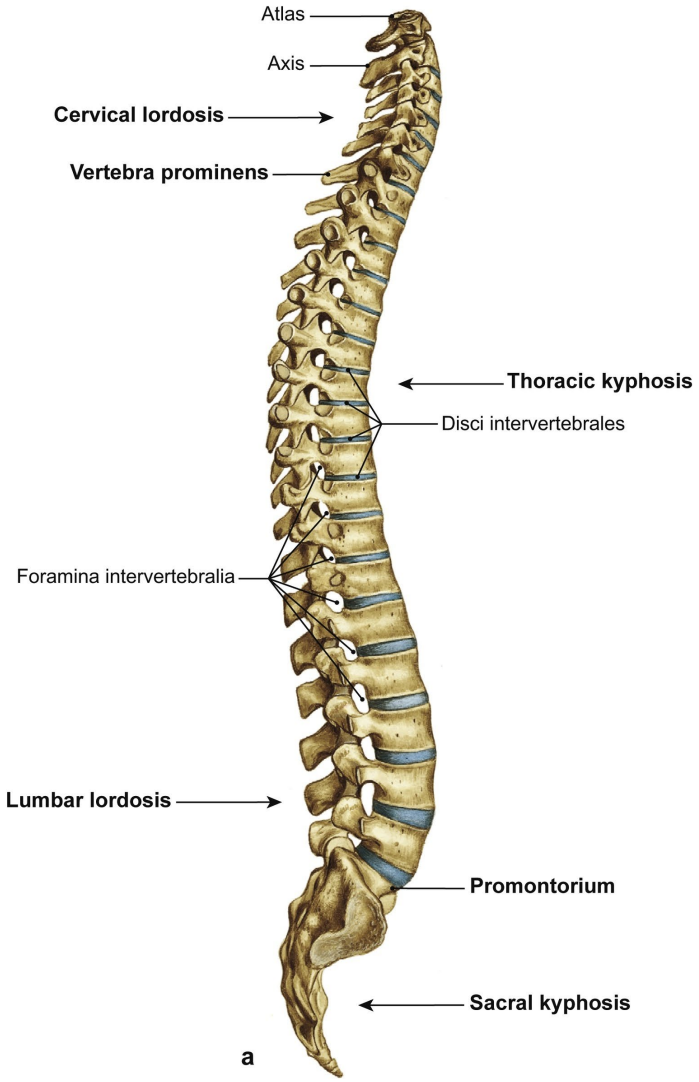
Hvis den femte lændehvirvel (L5) vokser sammen med korsbenet (os sacrum), så der kun er 23 frie hvirvler, kaldes det sakralisering.
If the fifth lumbar vertebra (L5) fuses with the sacrum (os sacrum), leaving only 23 separate vertebrae, this is called sacralization.

Lumbosakrale overgangshvirvler (sakralisering af L5 eller lumbalisering af S1) forekommer hos ca. 15 % af befolkningen.
Lumbosacral transitional vertebrae (sacralization of L5 or lumbarization of S1) occur in approximately 15% of the population.

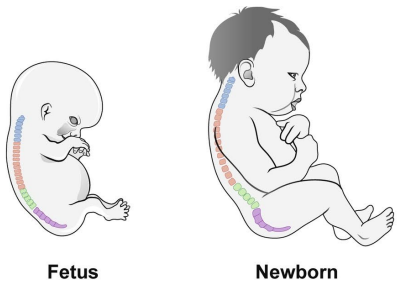


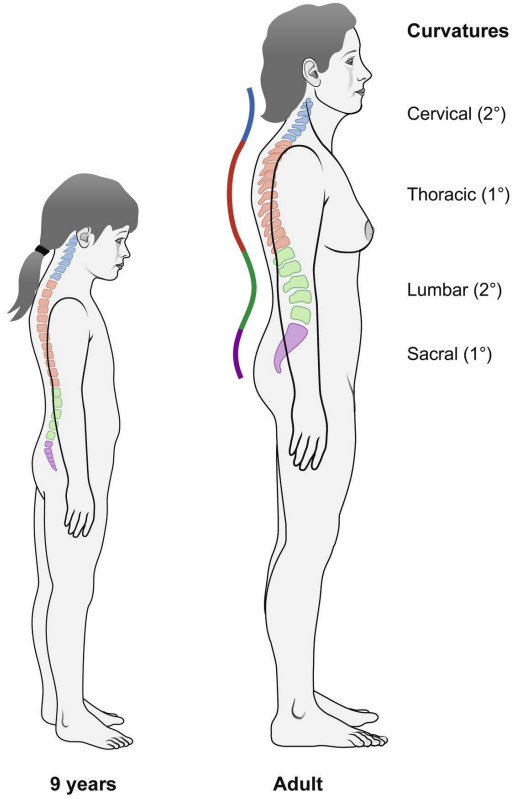
Rygsøjlens fysiologiske krumninger

Physiological Curvatures of the Vertebral Column



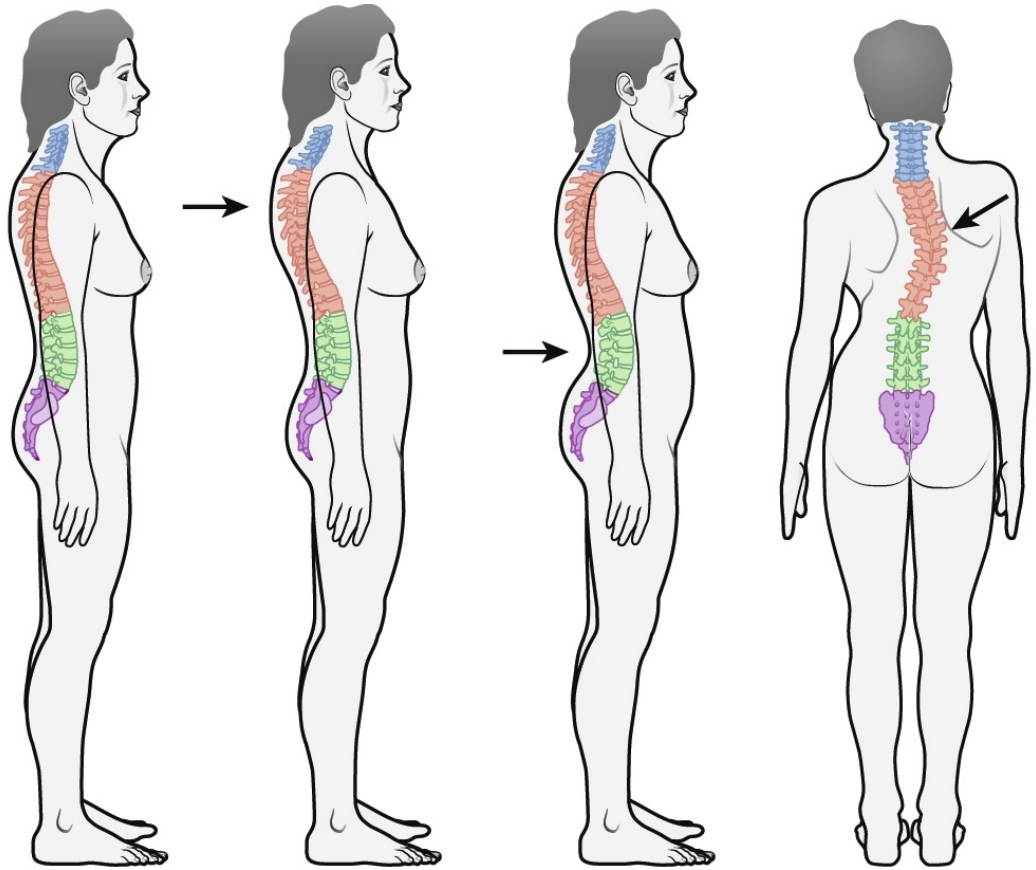
- Cervikal lordose: Krumning konveks fremad.
  Cervical lordosis: Convex anteriorly.
- Thorakal kyfose: Krumning konveks bagud.
  Thoracic kyphosis: Convex posteriorly.
- Lumbal lordose: Krumning konveks fremad.
  Lumbar lordosis: Convex anteriorly.
- Sakral kyfose: Krumning konveks bagud.
  Sacral kyphosis: Convex posteriorly.
- Halebenet (os coccygis): Består af op til fem sammenvoksede hvirvler og fortsætter den sakrale kyfose.
  Coccyx: Consists of up to five fused vertebrae and continues the sacral kyphosis.





Rygsøjlens fysiologiske krumninger

Physiological Curvatures of the Vertebral Column



Ryggens overfladeanatomi
Surface Anatomy of the Back

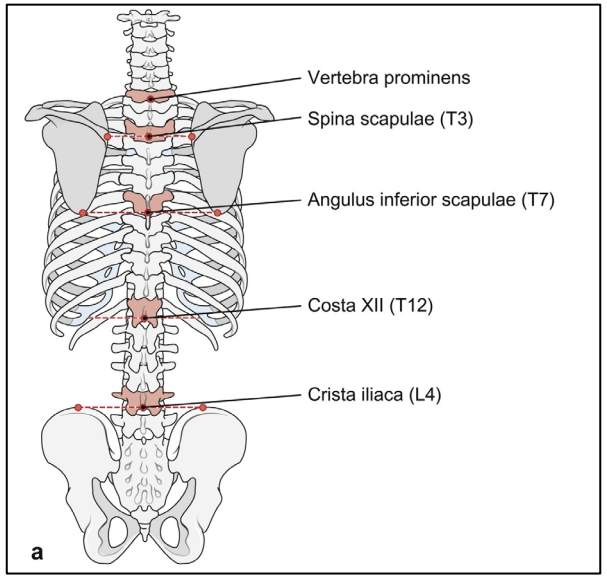
- C7 – Vertebra prominens: Tydeligt palpabel torntap ved nakkens basis.
  Prominent spinous process at the base of the neck.

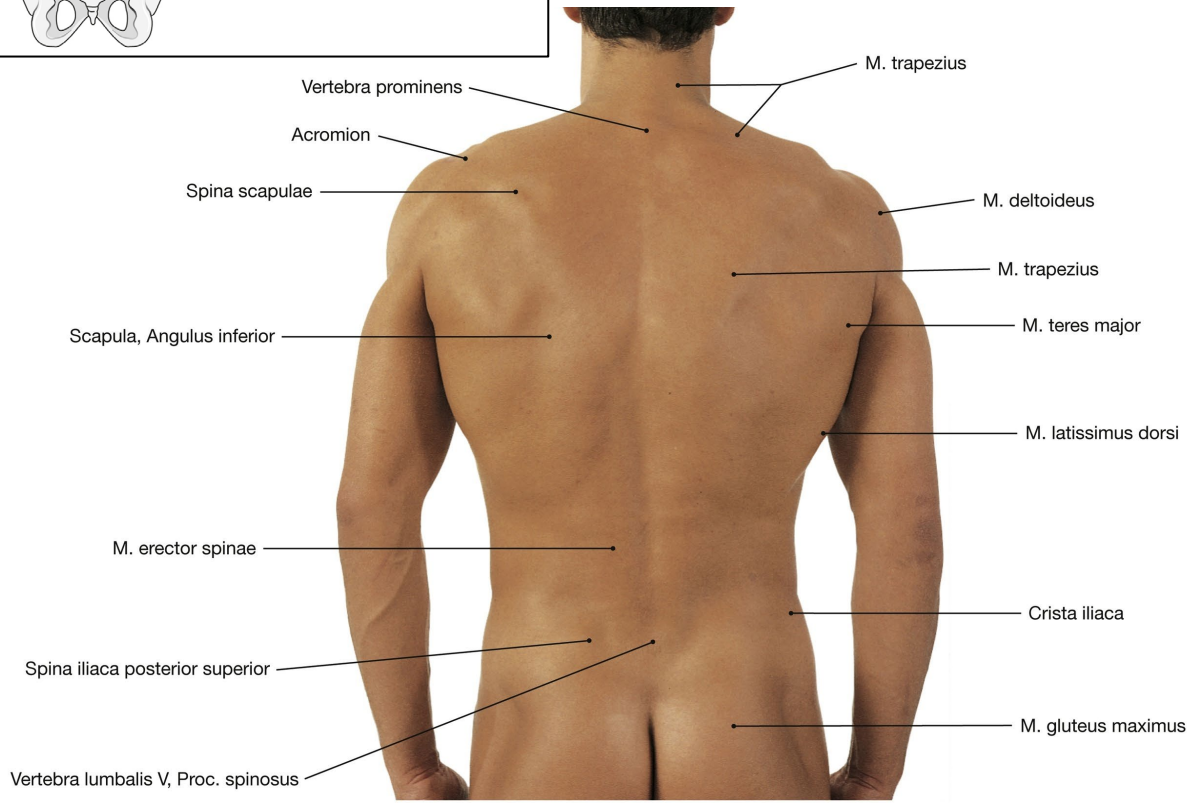
- T3 – Spina scapulae: Niveau for skulderbladets kam.
  Level of the spine of the scapula.

- T7 – Angulus inferior scapulae: Niveau for skulderbladets nedre vinkel.
  Level of the inferior angle of the scapula.

- T12 – Costa XII: Nederste ribben.
  Twelfth and lowest rib.

- L4 – Crista iliaca: Linjen mellem hoftekammene svarer omtrent til L4-niveau.
  The intercristal line approximately corresponds to the L4 level.

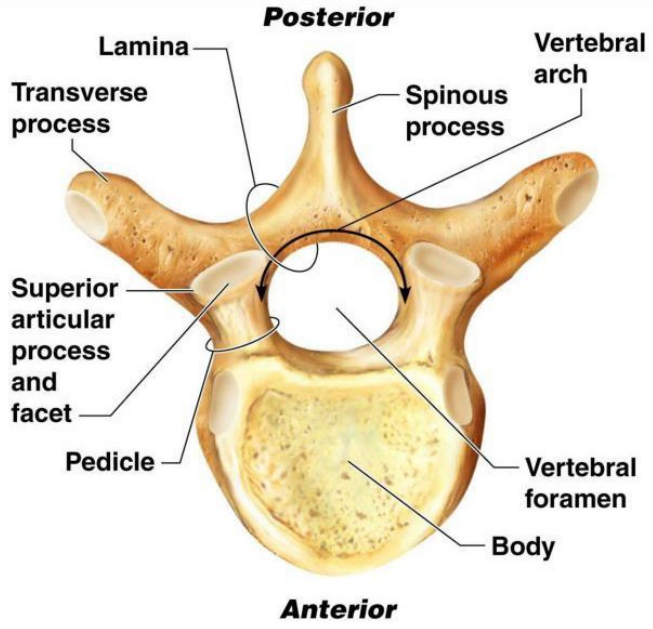


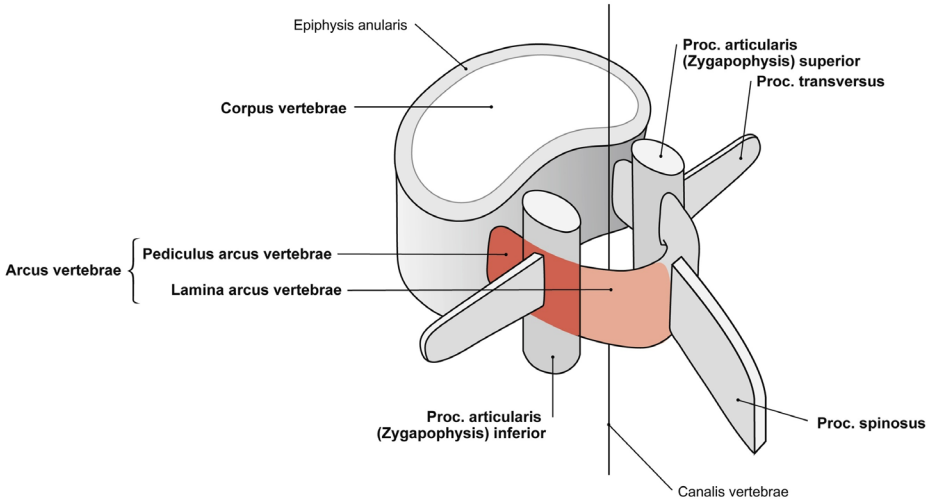


Typisk hvirvel – Anatomisk opbygning

Typical Vertebra – Anatomical Structure

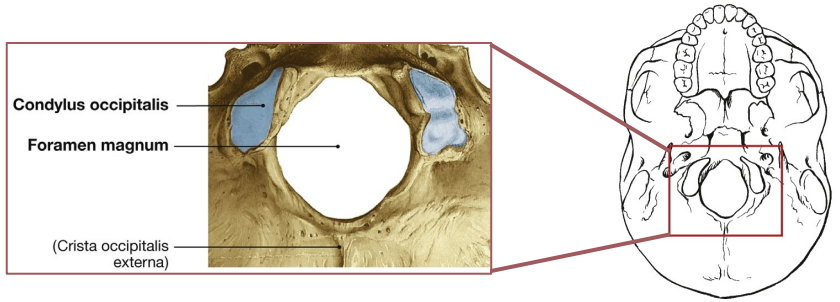
- Hvirvellegeme (***corpus vertebrae***): Vægtbæring.
  Vertebral body: Weight-bearing.
- Hvirvelbue (***arcus vertebrae***): Består af pediculi og laminae.
  Vertebral arch: Formed by pedicles and laminae.
- Hvirvelhul (***foramen vertebrale***): Bidrager til rygmarvskanalen.
  Vertebral foramen: Contributes to the vertebral canal.
- Tvær- og torntappe (***processus transversi et spinosus***): Hæfte for muskler og ligamenter.
- *Transverse and spinous processes: Attachment sites for muscles and ligaments.*
- Ledtappe (***processus articulares superiores et inferiores***): Danner facetled mellem hvirvlerne.
- *Articular processes: Form facet joints between vertebrae.*

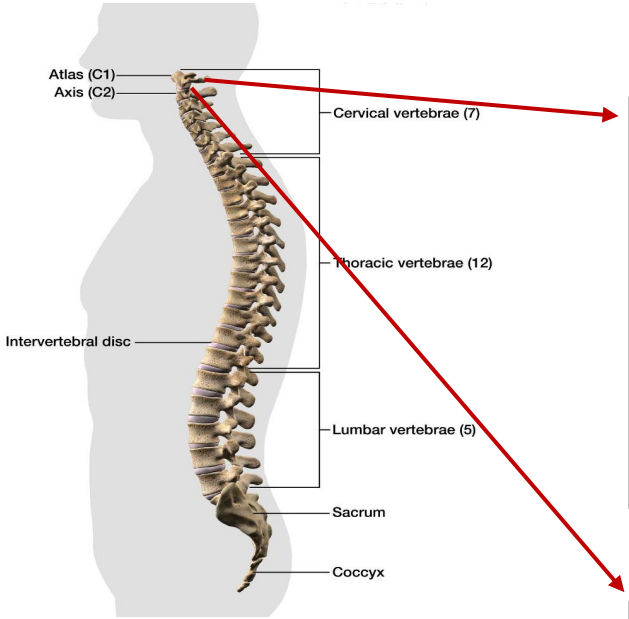




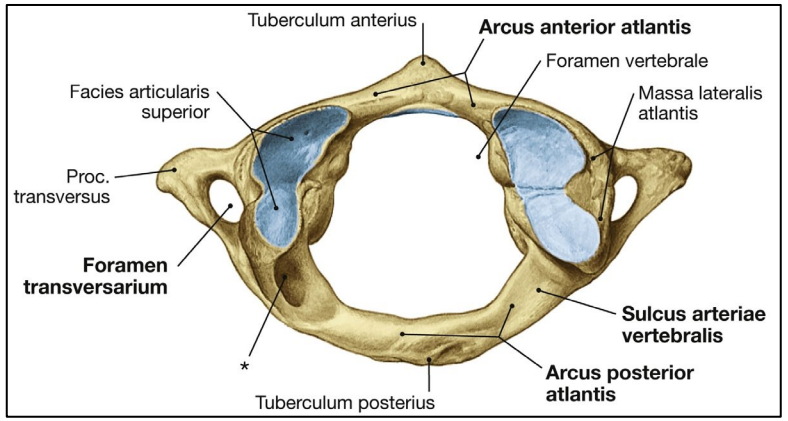
Særlige halshvirvler – Atlas (C1) og Axis (C2)

Specialized Cervical Vertebrae – Atlas (C1) and Axis (C2)

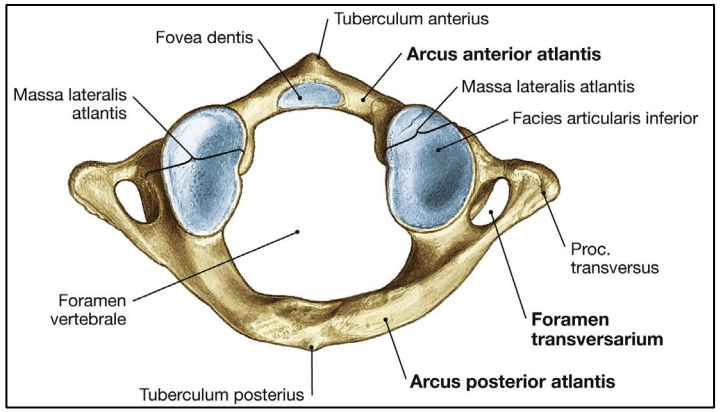




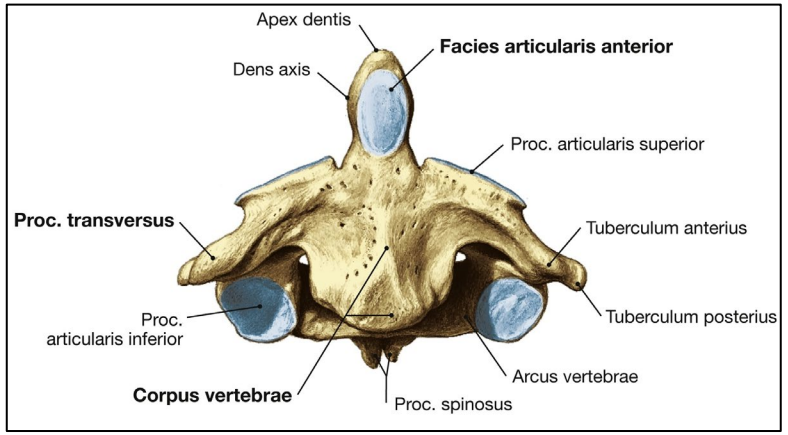
C1, atlas; Cranial view



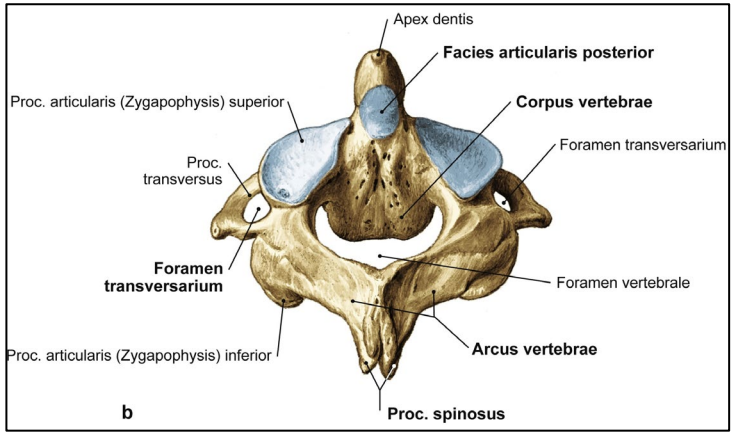
C1, atlas; Caudal view



C2, axis; Ventral view.

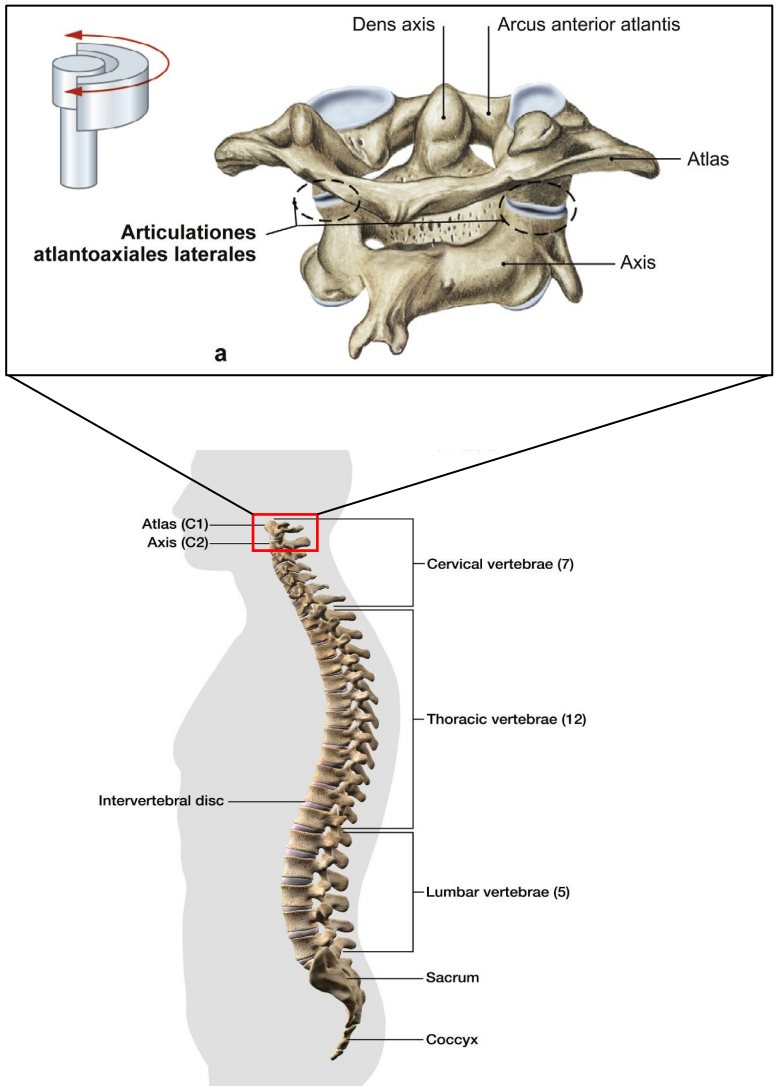


C2, Dorsocranial view



Særlige halshvirvler – Atlas (C1) og Axis (C2)

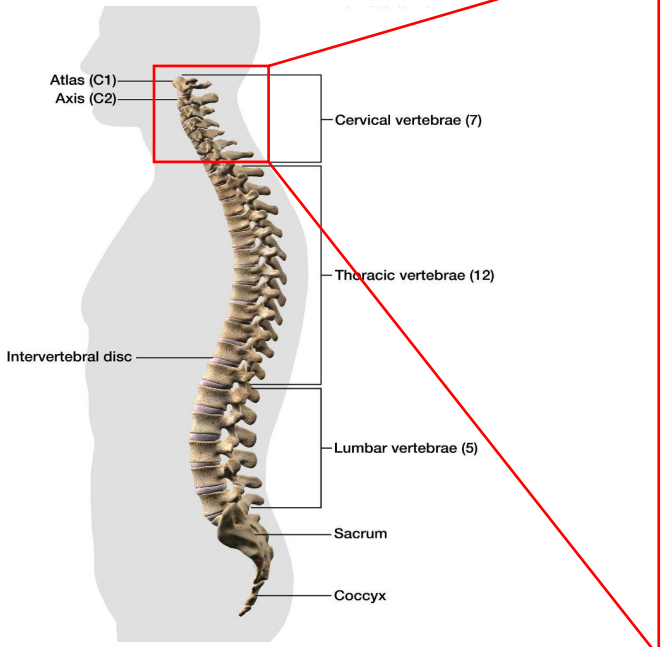
Specialized Cervical Vertebrae – Atlas (C1) and Axis (C2)

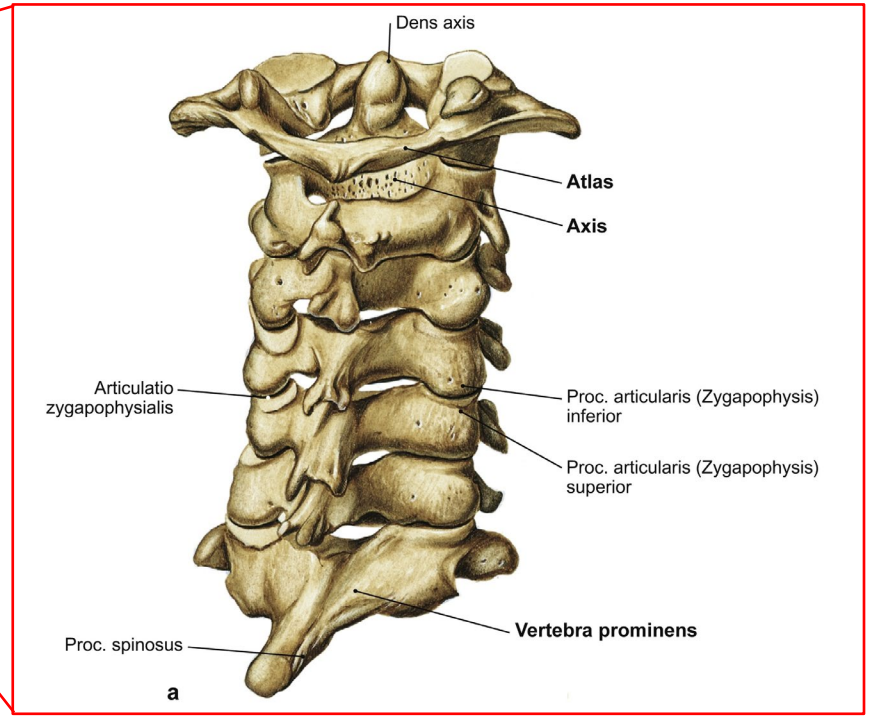


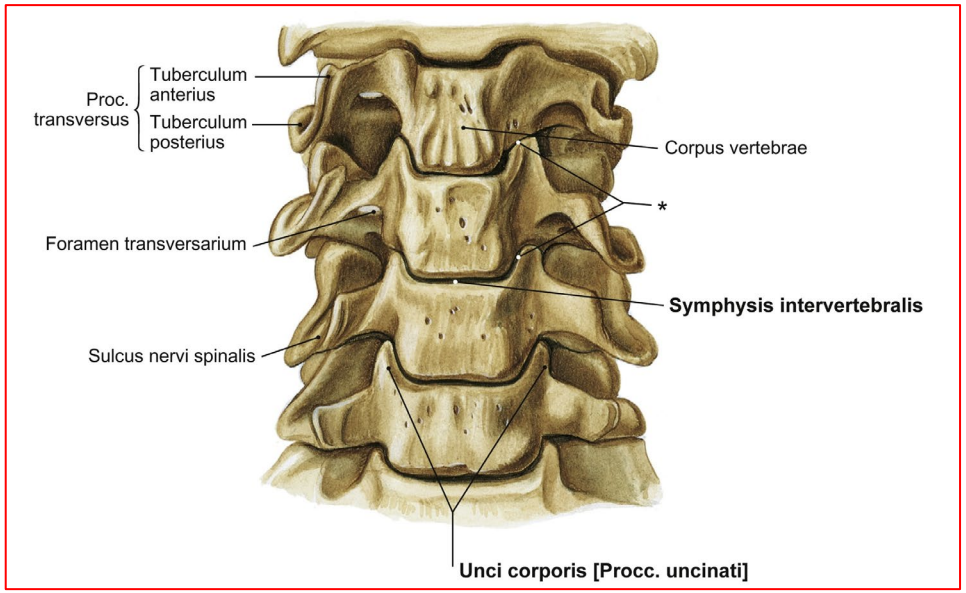


Halshvirvler

Cervical Vertebrae







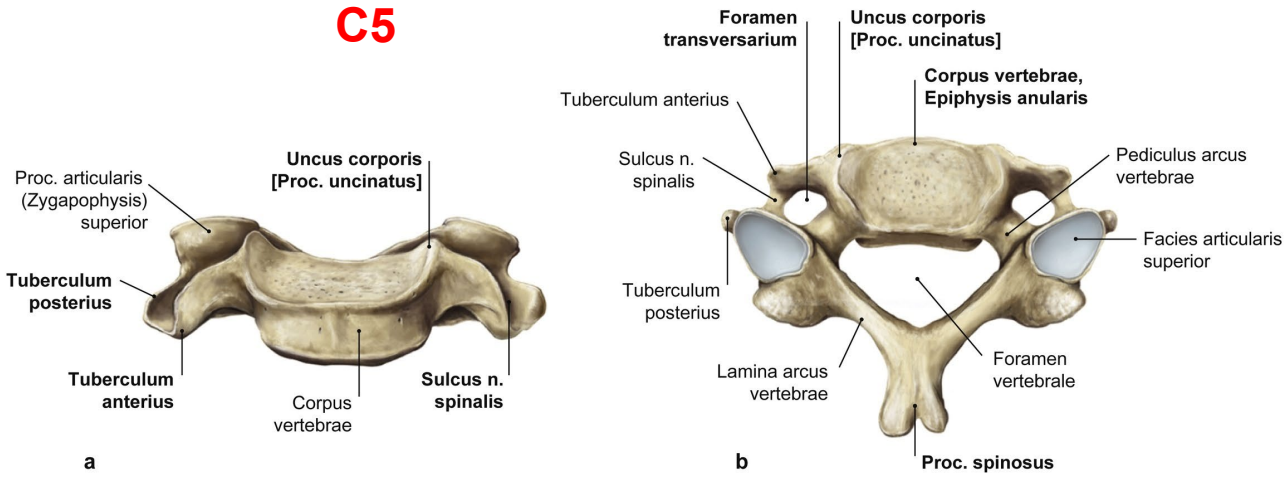
Typisk halshvirvel (C3–C6)

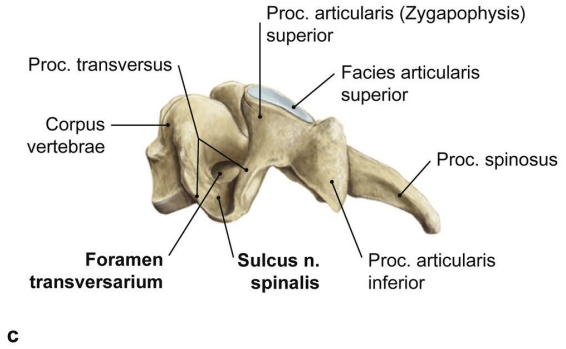
Typical Cervical Vertebra (C3–C6)

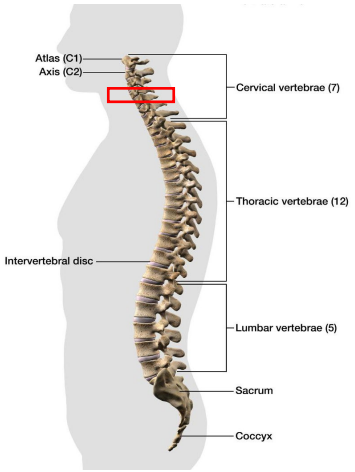
- Lille hvirvellegeme (*corpus vertebrae*)
  Small vertebral body
- Har foramen transversarium i processus transversus
  Contains a transverse foramen in the transverse process
- Torntappen (processus spinosus) er ofte kort og todelt
  The spinous process is often short and bifid
- Hvirvelhullet (foramen vertebrale) er stort og trekantet
  The vertebral foramen is large and triangular

Velegnet til stor bevægelighed i nakken
Adapted for high mobility in the neck

Halscolumna prioriterer bevægelighed frem for vægtbæring
The cervical spine prioritizes mobility over weight-bearing



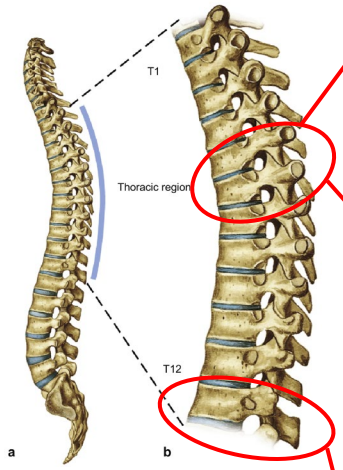


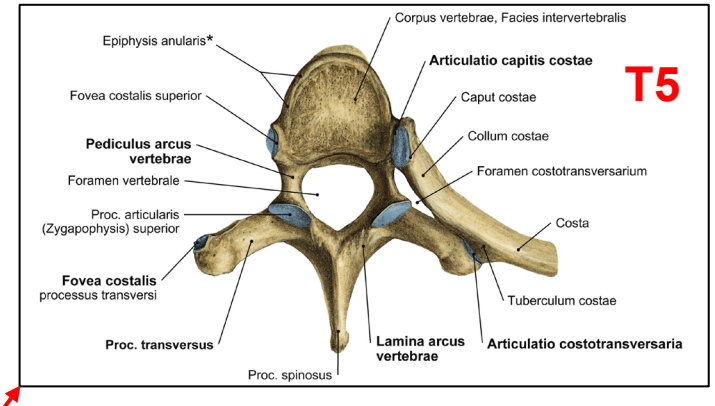


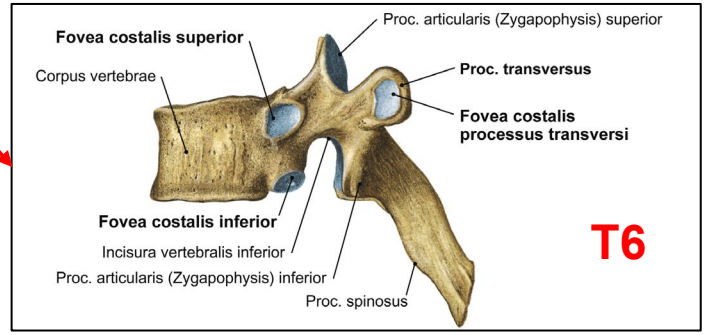
Typisk thorakalhvirvel (T2–T8)
Typical Thoracic Vertebra (T2–T8)

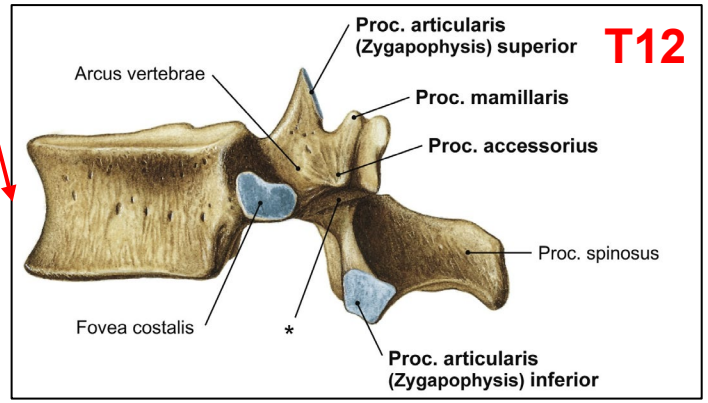
- **Mellemstort hvirvellegeme (*corpus vertebrae*)**
  Medium-sized vertebral body
- **Har ledflader til ribbenene (*foveae costales*)**
  Has costal facets for articulation with ribs
- **Torntappen (*processus spinosus*) er lang og hælder nedad**
  The spinous process is long and slopes downward
- **Hvirvelhullet (*foramen vertebrale*) er relativt lille og rundt**
  The vertebral foramen is relatively small and round
- **Stabiliseres af brystkassen og tillader mindre bevægelighed**
  Stabilized by the rib cage and allows less mobility

Thorakalcolumna balancerer stabilitet og respiration
The thoracic spine balances stability and respiration









Typisk lumbalhvirvel (L1–L4)

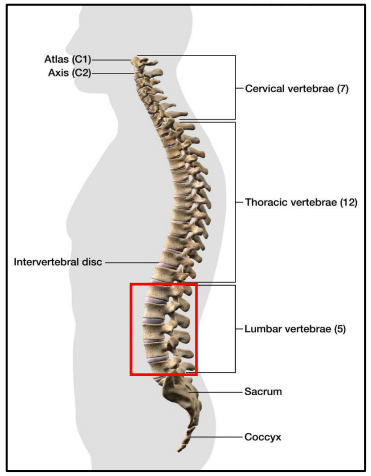
Typical Lumbar Vertebra (L1–L4)

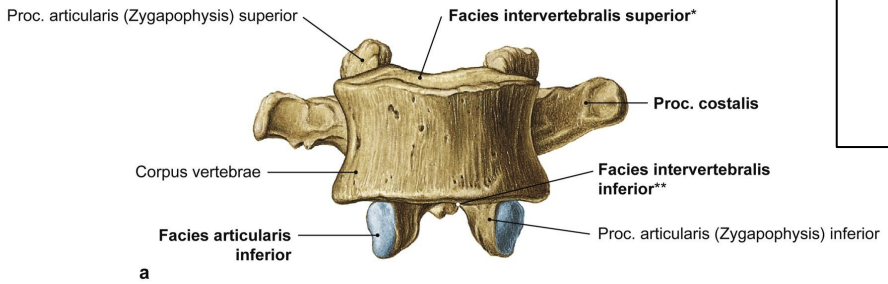
- Hvirvellegeme (*corpus vertebrae*): Stort og kraftigt til vægtbæring.
  Vertebral body: Large and robust for weight-bearing.

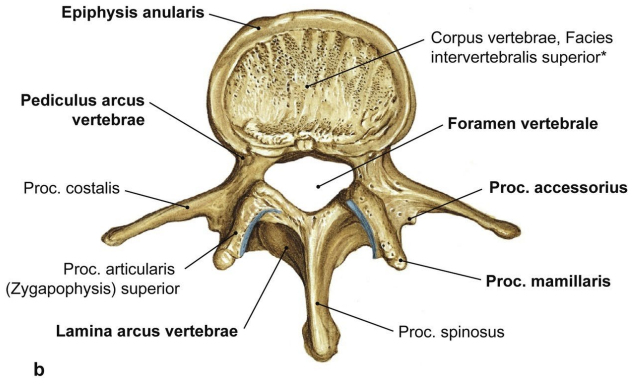
- Kraftig torntap (*processus spinosus*): Kort, bred og næsten vandret.
  Robust spinous process: Short, broad, and nearly horizontal.

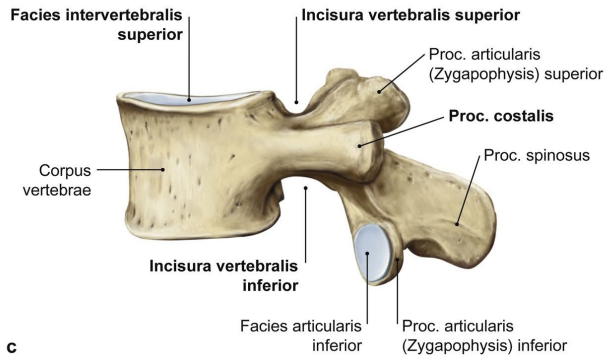
- Særlige knoglefremspring (processus costales, mamillares et accessorii): Karakteristiske for lændehvirvlerne.
  Costal, mammillary, and accessory processes: Characteristic landmarks of lumbar vertebrae.

- Facetled (articulationes zygapophysiales): Overvejende orienteret i sagittalplanet – tillader fleksion og ekstension, men begrænser rotation.
  Facet joints: Predominantly sagittally oriented, favoring flexion and extension while limiting rotation.





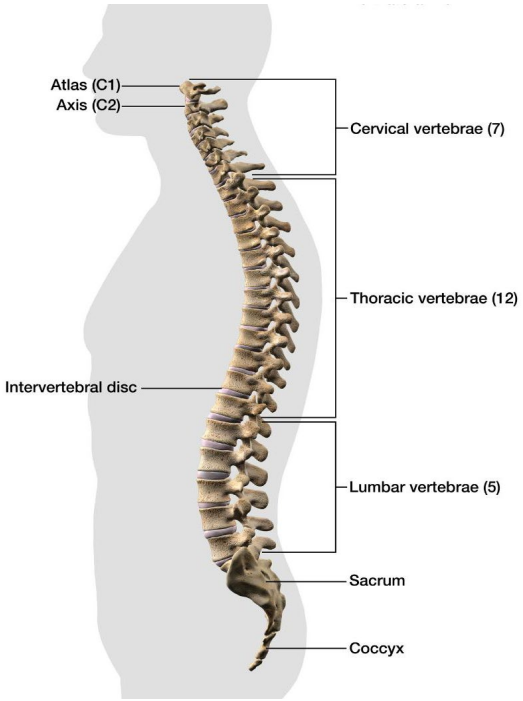




Lændehvirvlerne er tilpasset vægtbæring og fleksion/ekstension.
Lumbar vertebrae are adapted for weight-bearing and flexion/extension.

Typisk hvirvel – Anatomisk opbygning

Typical Vertebra – Anatomical Structure



- Halshvirvler (Cervikal): Lille hvirvellegeme, stor bevægelighed og foramen transversarium.
  Cervical vertebrae: Small vertebral body, high mobility, and transverse foramen.

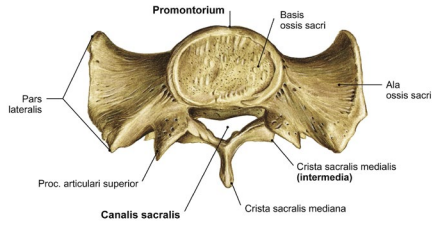
- Brysthvirvler (Thorakal): Ledforbindelser med ribbenene, stor stabilitet og lange torntappe.
  Thoracic vertebrae: Rib articulations, high stability, and long spinous processes.

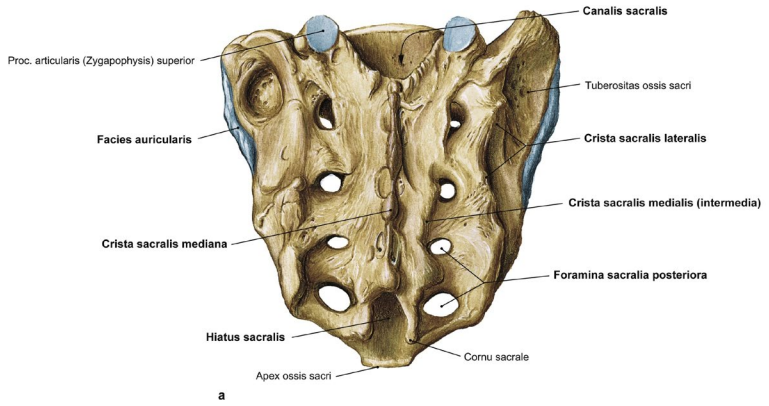
- Lændehvirvler (Lumbal): Stort hvirvellegeme, kraftige knogletappe og tilpasset vægtbæring.
  Lumbar vertebrae: Large vertebral body, robust processes, and adapted for weight-bearing.

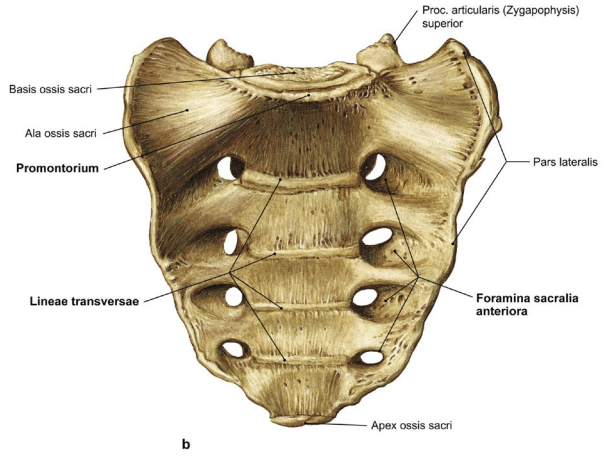
Korsbenet (*os sacrum*)

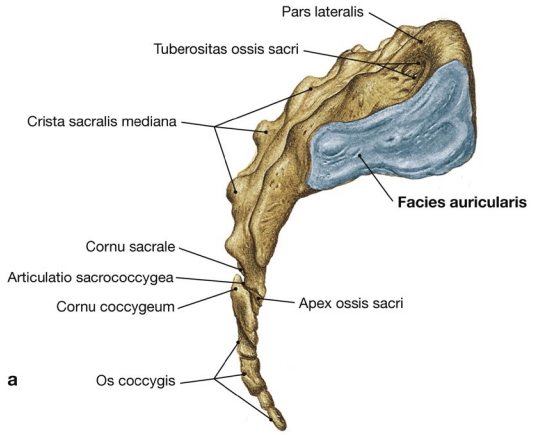
Sacrum

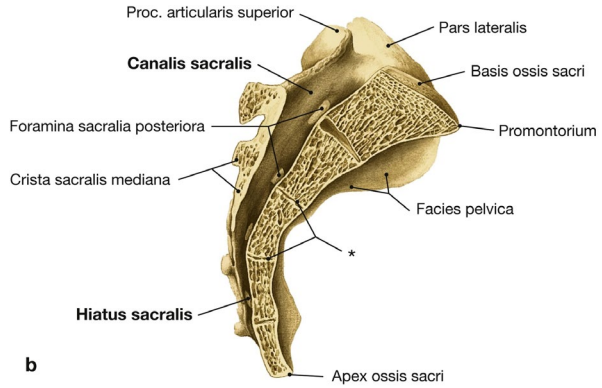
- Fem sammenvoksede hvirvler: overfører kropsvægten til bækkenet.
  Five fused vertebrae: transfer body weight to the pelvis.
- Sakralkamme (***cristae sacrales***): Sammenvoksede hvirveltappe.
  Sacral crests: Fused vertebral processes.
- Sakralhuller (***foramina sacralia***): Passage for sakrale nervegrene.
  Sacral foramina: Openings for sacral nerve branches.
- Sakralhiatus (hiatus sacralis): Åbning til sakralkanalen; epidural anæstesi.
  Sacral hiatus: Opening of the sacral canal; epidural anesthesia.
- Sakralpromontorium (promontorium): Forreste fremspring på S1.
  Sacral promontory: Anterior projection of S1.











Sakroiliakalleddet (art. sacroiliaca)

Sacroiliac joint

Led mellem os sacrum og os ilium

Joint between the sacrum and ilium

Stærke ligamenter giver stabilitet

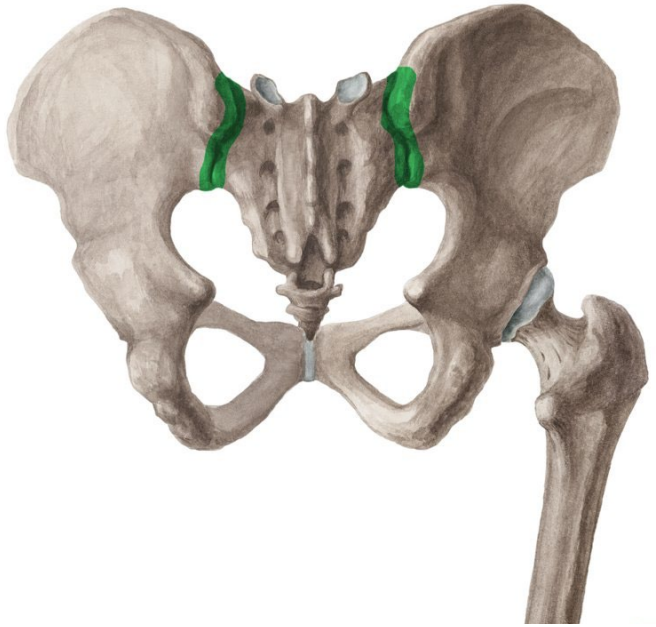
Strong ligaments provide stability

Overfører belastning fra columna til bækkenet

Transfers load from the spine to the pelvis

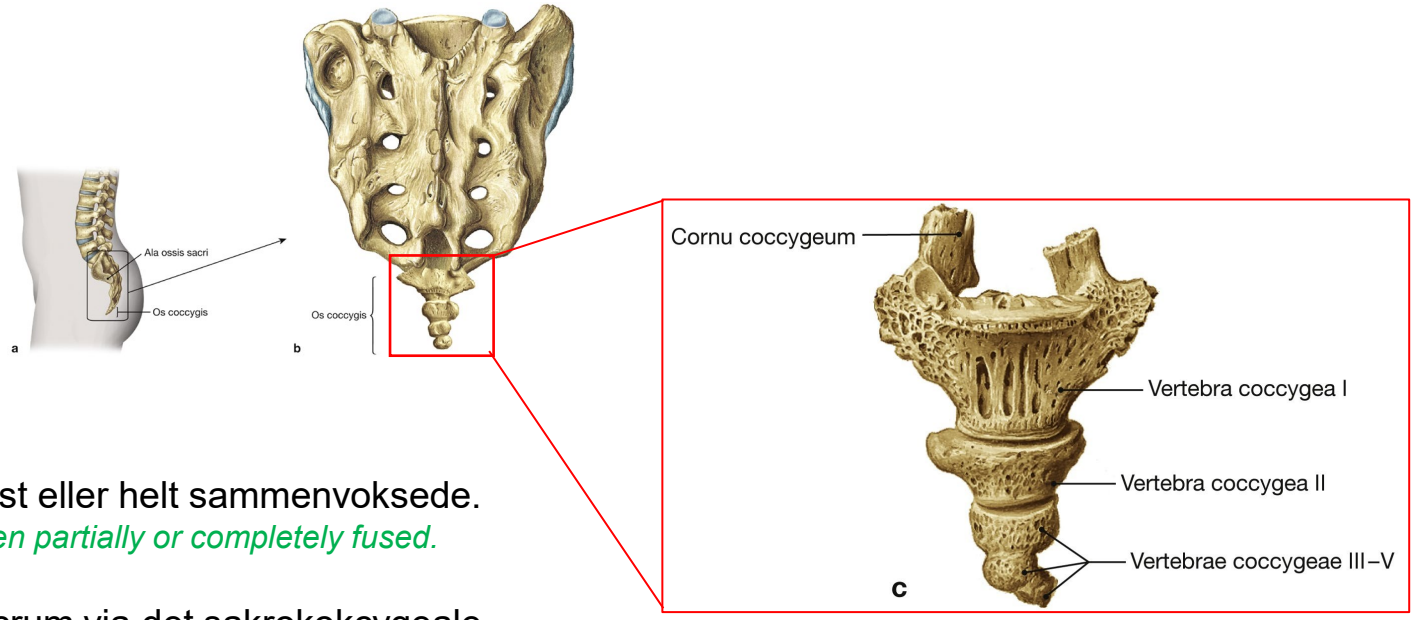
Kun små bevægelser (nutation og counternutation)

Only small movements (nutation and counternutation)



Halebenet (*os coccygis*)

Coccyx



- 3–5 rudimentære hvirvler: Typisk 3–4, ofte delvist eller helt sammenvoksede.
  Usually 3–4 rudimentary vertebrae, sometimes five, often partially or completely fused.
- Forbindelse til korsbenet: Forbindes med os sacrum via det sakrokokcygeale led og cornua coccygea.
  Articulates with the sacrum through the sacrococcygeal joint, with the coccygeal cornua contributing to the connection.
- Aftagende størrelse: Hvirvlerne bliver mindre i kaudal retning; den første er mest udviklet.
  Vertebrae decrease in size inferiorly; the first coccygeal vertebra is the most developed.

- Typiske skader: Kontusion, luksation eller fraktur.
  Common injuries: Contusion, dislocation, or fracture.
- Årsager: Fald på halebenet eller gentagen belastning, fx cykling og roning.
  Causes: Falls onto the tailbone or repetitive stress, e.g., cycling and rowing.

Læringsmål – Systematisk anatomi

Learning Objectives – Systematic Anatomy

- **Beskrive rygsøjlens regioner, hvirvler og fysiologiske krumninger.**

  Describe spinal regions, vertebral morphology, and physiological curves.

- **Forklare facetled, ligamenter og disci samt deres funktion.**

  Explain facet joints, ligaments, discs and their functions.

- **Identificere overfladiske og dybe rygmuskler og deres hovedfunktion.**

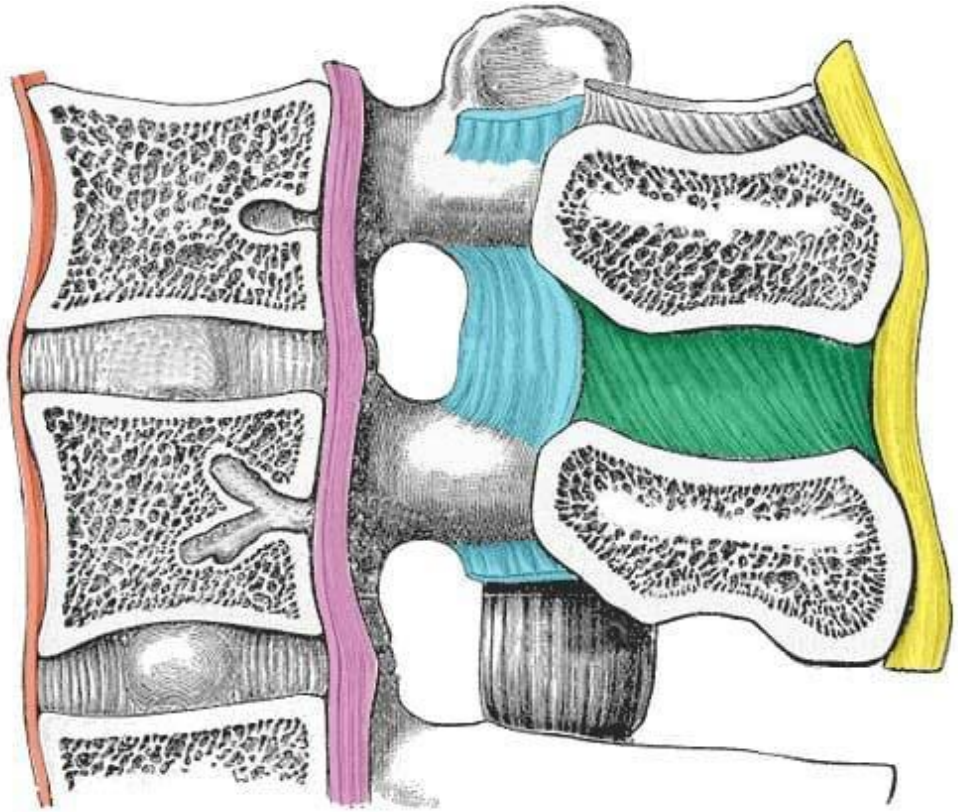
  Identify major extrinsic/intrinsic back muscles and their actions.

- **Forklare fleksion, ekstension, lateralfleksion og rotation.**

  Explain spinal flexion, extension, lateral flexion and rotation.

Rygsøjlens vigtigste ligamenter

Major Ligaments of the Vertebral Column



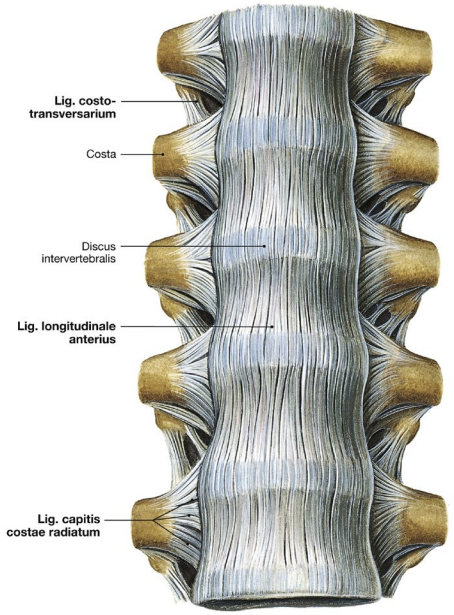
Anterior longit. ligament
Posterior longit. ligament
Ligamentum flavum
Interspinal ligament
Supraspinous ligament

© TeachMeAnatomy

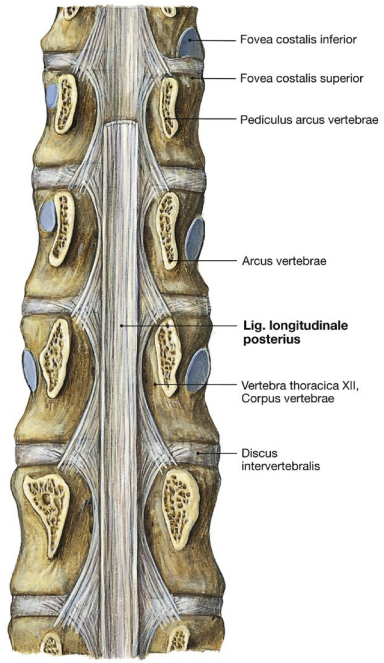
Rygsøjlens vigtigste ligamenter

Major Ligaments of the Vertebral Column

ventral view



dorsal view



Forreste længdeligament (***lig. longitudinale anterius***): Begrænser ekstension.

Anterior longitudinal ligament: Limits extension.

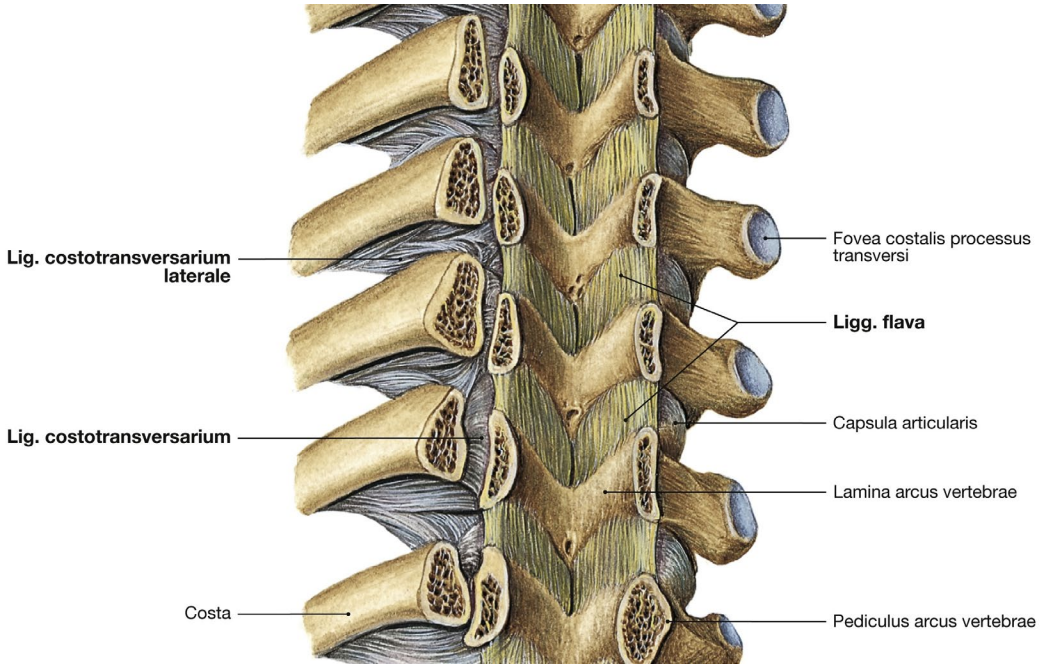
Bageste længdeligament (***lig. longitudinale posterius***): Begrænser fleksion.

Posterior longitudinal ligament: Limits flexion.

Funktion: Stabiliserer rygsøjlen og understøtter de intervertebrale disci.

Function: Stabilize the spine and support the intervertebral discs.

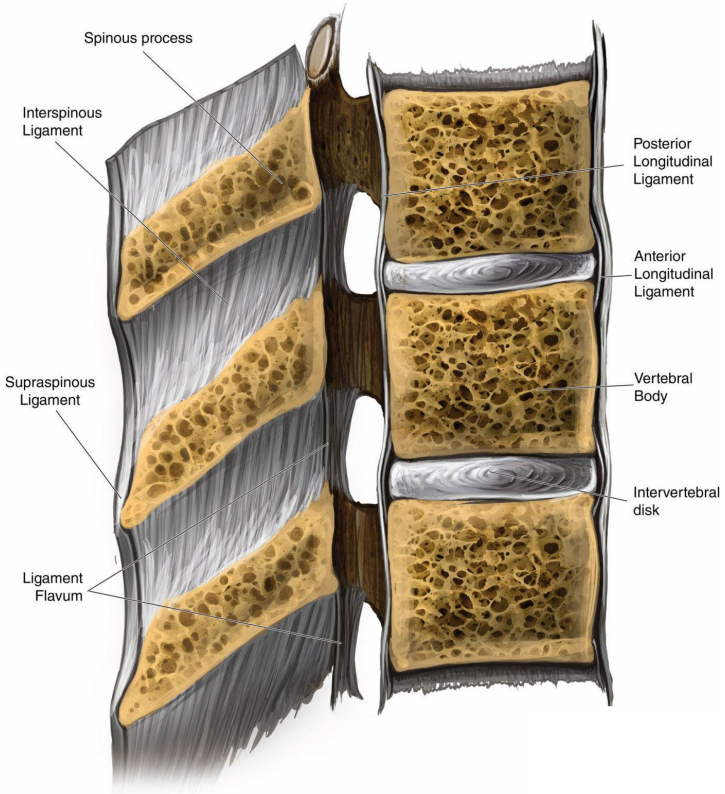
Rygsøjlens vigtigste ligamenter
Major Ligaments of the Vertebral Column



Gule ligamenter (***ligamenta flava***)
Yellow Ligaments

- Forbinder laminae på tilstødende hvirvler og er rige på elastiske fibre.
  Connect the laminae of adjacent vertebrae and contain abundant elastic fibers.
- Danner en del af spinalkanalens bagvæg.
  Form part of the posterior wall of the vertebral canal.
- Funktion: Modvirker fleksion og hjælper rygsøjlen tilbage til oprejst stilling.
  Function: Resist flexion and assist the return to an upright posture through elastic recoil.

Rygsøjlens vigtigste ligamenter
Major Ligaments of the Vertebral Column



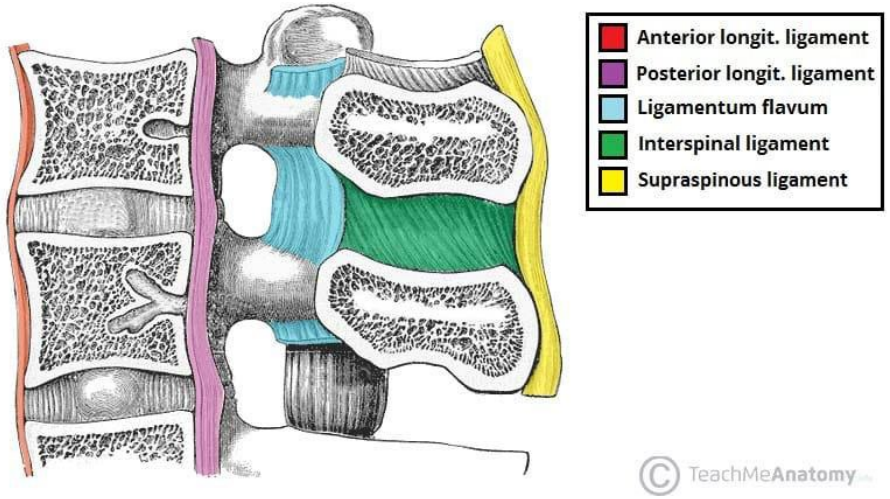
- Interspinøse ligamenter (***ligg. interspinalia***): Forbinder tilstødende torntappe.
  Interspinous ligaments: Connect adjacent spinous processes.

- Supraspinøst ligament (***lig. supraspinale***): Forbinder torntappenes spidser.
  Supraspinous ligament: Connects the tips of spinous processes.

Funktion: Modvirker overdreven fleksion og bidrager til stabilitet.
Function: Resist excessive flexion and contribute to stability.

Rygsøjlens vigtigste ligamenter

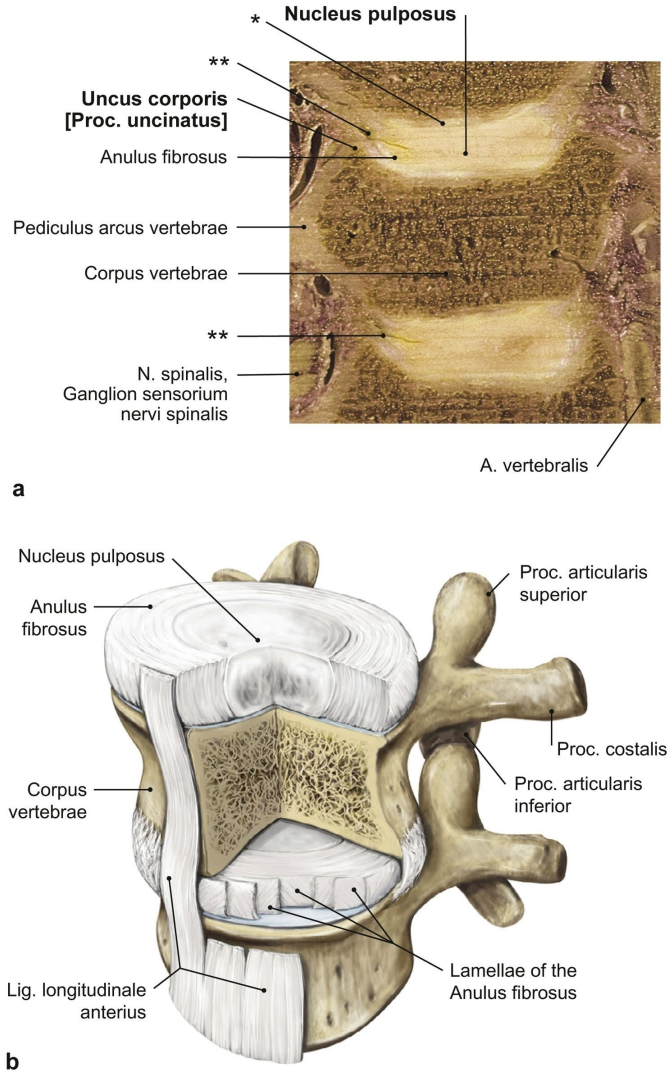
Major Ligaments of the Vertebral Column



Ligament / Ligament	Forbinder / Connects	Hovedfunktion / Main function
Lig. longitudinale anterius	Forsiden af hvirvellegemer og disci. *Anterior surfaces of vertebral bodies and discs.*	Begrænser ekstension. *Limits extension (backward bending).*
Lig. longitudinale posterius	Bagsiden af hvirvellegemer og disci, inde i spinalkanalen. *Posterior surfaces of vertebral bodies and discs, within the vertebral canal.*	Begrænser fleksion. *Limits flexion (forward bending).*
Ligg. flava	Hvirvelbuerne (laminae) på tilstødende hvirvler. *Laminae of adjacent vertebrae.*	Modvirker fleksion; elastisk tilbagespring hjælper med at rette ryggen op. *Resist flexion; elastic recoil assists return to upright posture.*
Ligg. interspinalia	Mellem tilstødende torntappe. *Between adjacent spinous processes.*	Begrænser fleksion. *Limit flexion.*
Lig. supraspinale	Langs spidserne af torntappene. *Along the tips of spinous processes.*	Begrænser fleksion. *Limits flexion.*

Mellemhvirvelskivens opbygning

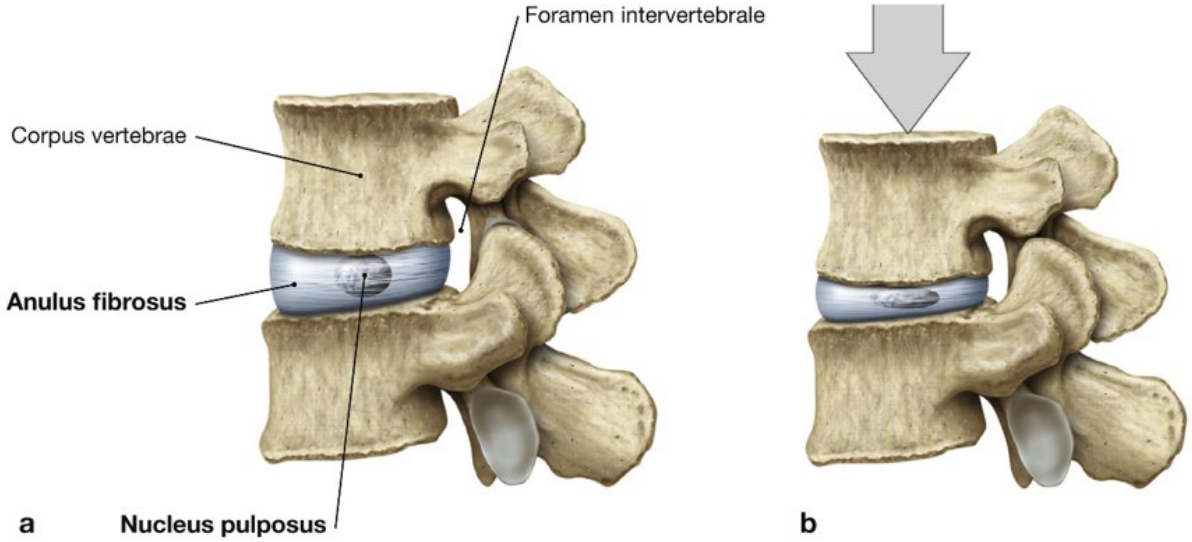
Structure of the Intervertebral Disc



- Fiberring (***anulus fibrosus***): Ydre ring af kollagenfibre, som modstår belastning.
  Annulus fibrosus: Outer collagen-rich ring that resists mechanical loading.
- Gelékerne (***nucleus pulposus***): Væskerig kerne, som fordeler trykbelastning.
  Nucleus pulposus: Water-rich core that distributes compressive loads.
- Endeplader: Forbinder discus med tilstødende hvirvellegemer.
  Endplates: Form the interface between the disc and adjacent vertebral bodies.
- Funktion: Muliggør bevægelse og fordeler belastningen mellem hvirvlerne.
  Function: Permit movement and distribute loads between vertebrae.

Mellemhvirvelskivens function

Function of the Intervertebral Disc



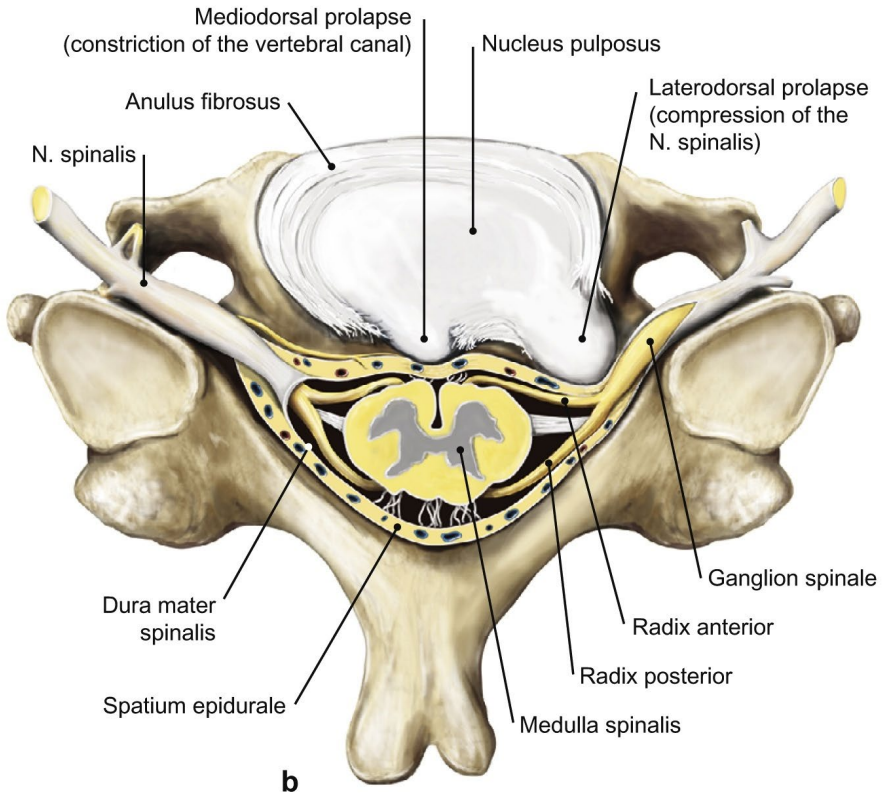
- Støddæmpning: Disci intervertebrales fordeler tryk og mekanisk belastning.
Shock absorption: Intervertebral discs distribute pressure and mechanical loads.

- Bevægelighed: Tillader små bevægelser mellem tilstødende hvirvler.
Mobility: Allow small movements between adjacent vertebrae.

- Væskeindhold: Discus mister væske under dagens belastning og genoptager væske under hvile.
Hydration: Discs lose fluid during daily loading and rehydrate during rest.

Diskusprolaps og nervekompression

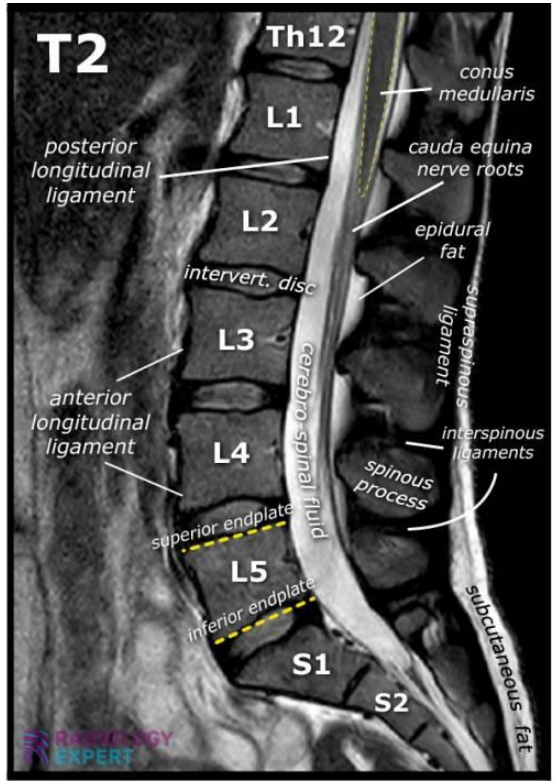
Disc Herniation and Nerve Root Compression

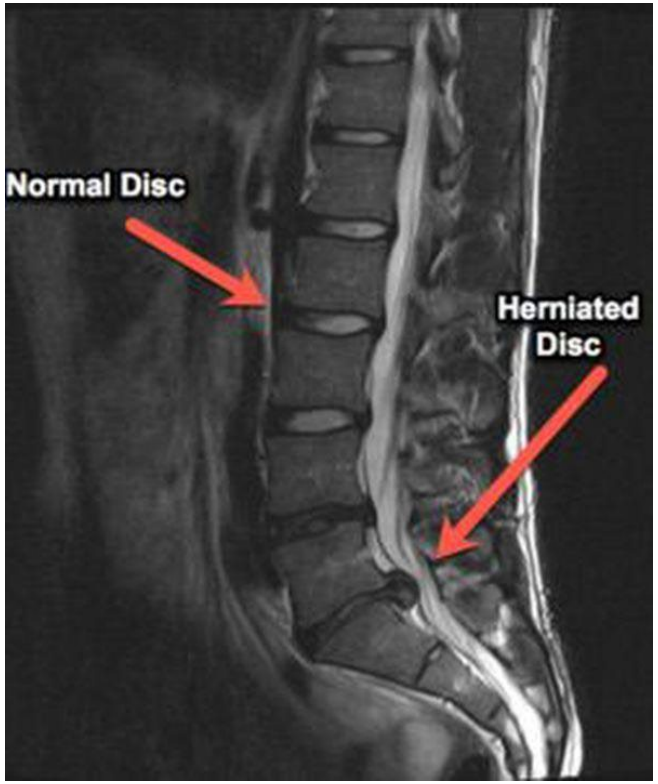


- Diskusprolaps: Discmateriale forskydes ud over discus' normale grænser.
  Disc herniation: Disc material extends beyond the normal disc boundaries.
- Posterolateral prolaps: Kan komprimere en spinalnerverod.
  Posterolateral herniation: May compress a spinal nerve root.
- Symptomer: Udstrålende smerter, føleforstyrrelser og muskelsvaghed.
  Symptoms: Radiating pain, sensory disturbances, and muscle weakness.
- Hyppige niveauer: L4–L5 og L5–S1.
  Common levels: L4–L5 and L5–S1.

Diskusprolaps – MR-skanning

Disc Herniation – MRI



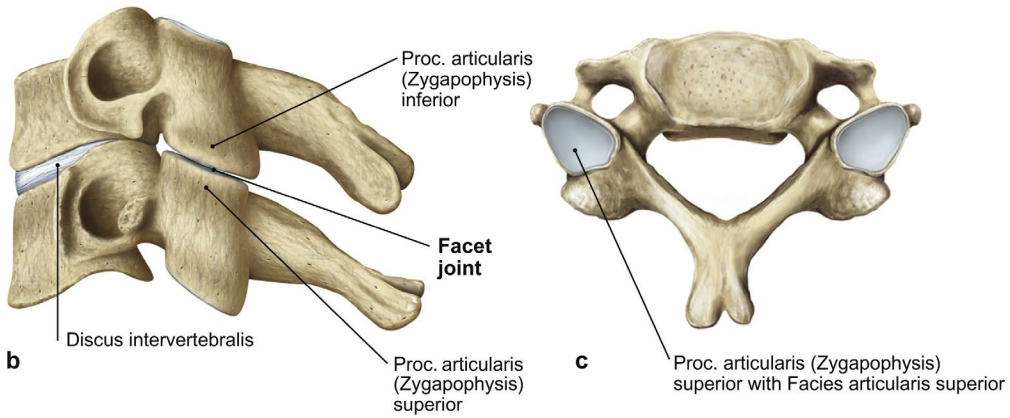


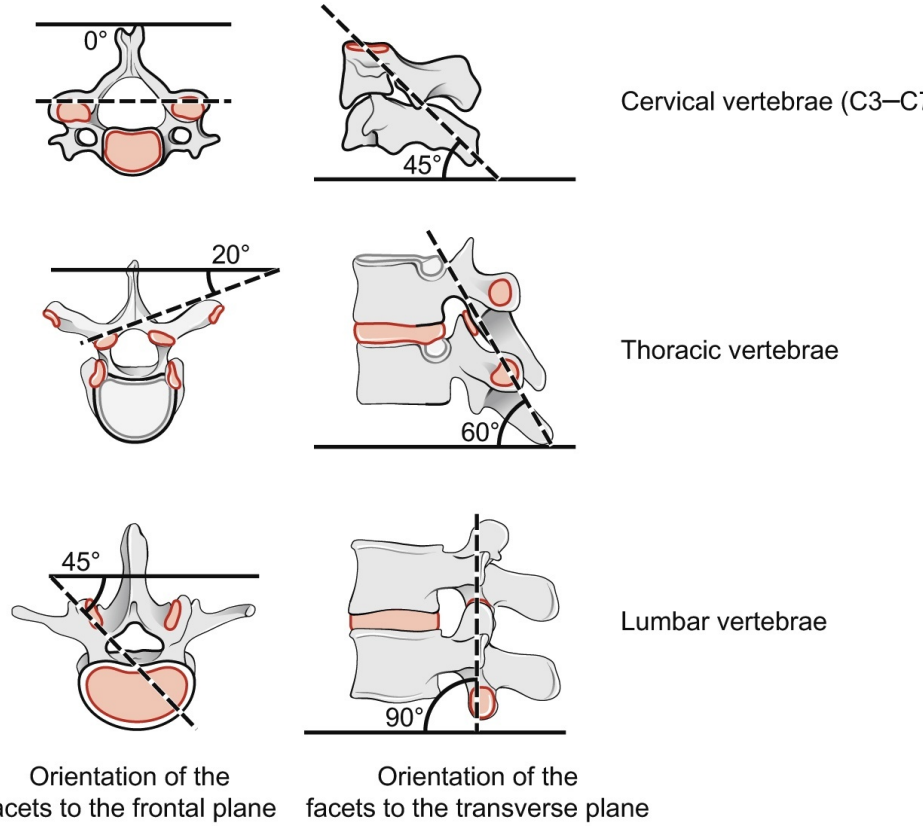
- Normal discus: Bevarer sin kontur uden fokal prolaps.
  Normal disc: Maintains its contour without focal herniation.
- Diskusprolaps: Discmateriale trænger ud over discus normale grænser.
  Disc herniation: Disc material extends beyond its normal boundaries.
- Klinisk betydning: Kan medføre nerverodskompression og radikulære symptomer.
  Clinical relevance: May compress nerve roots and cause radicular symptoms.

Facetled – Bevægelse og stabilitet

Facet Joints – Movement and Stability

- Ledtappe danner facetled (articulationes zygapophysiales).
  Articular processes form synovial facet joints.
- Ledfladernes orientering varierer mellem rygsøjlens regioner.
  Facet orientation varies across the cervical, thoracic and lumbar regions.
- Lænderyggens facetled fremmer fleksion/ekstension og begrænser rotation.
  Lumbar facets favor flexion/extension and restrict axial rotation.
- Facetled og disci bidrager sammen til stabilitet og belastningsfordeling.
  Facet joints and discs jointly contribute to stability and load distribution.





Læringsmål – Systematisk anatomi

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  Explain facet joints, ligaments, discs and their functions.

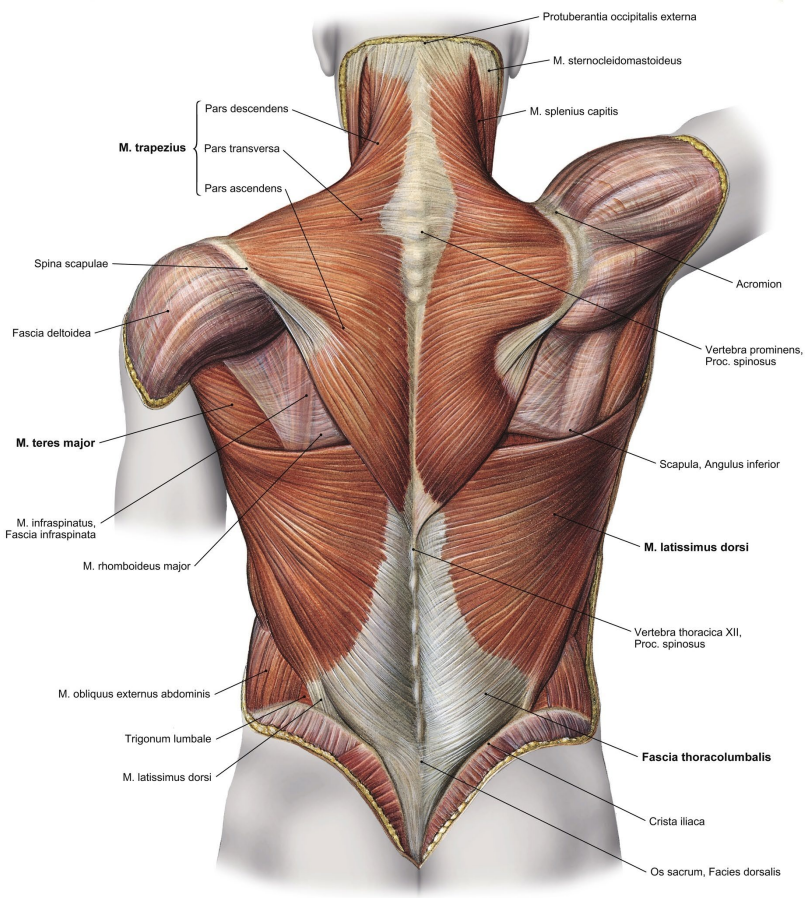
- **Identificere overfladiske og dybe rygmuskler og deres hovedfunktion.**

  Identify major extrinsic/intrinsic back muscles and their actions.

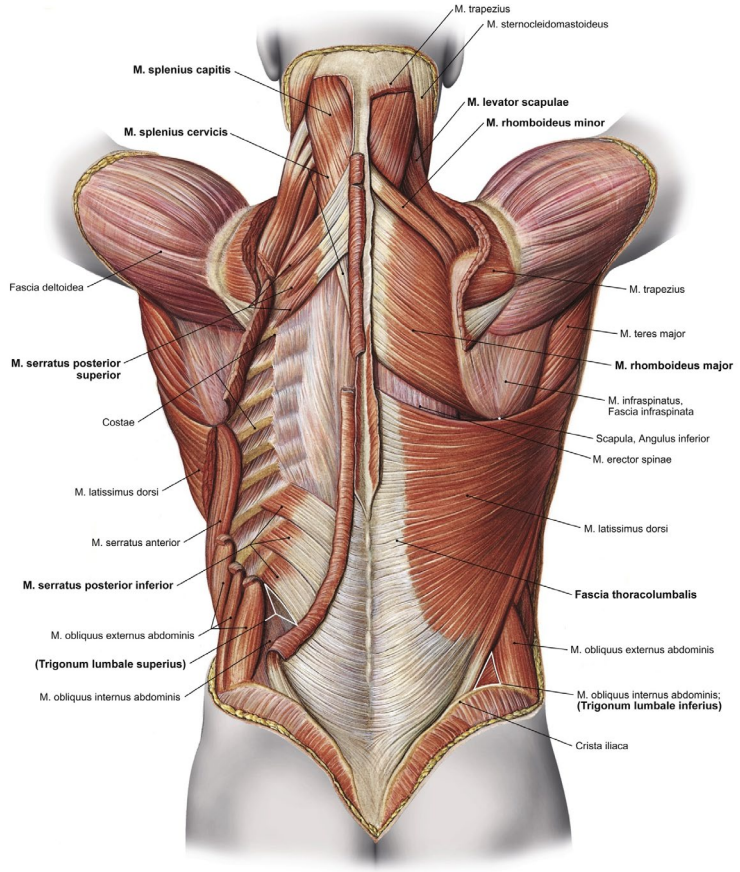
- **Forklare fleksion, ekstension, lateralfleksion og rotation.**

  Explain spinal flexion, extension, lateral flexion and rotation.

Overfladiske rygmuskler
Superficial Back Muscles



Muskler mellem rygsøjlen og scapula
Muscles Connecting the Spine and Scapula



Trækker overarmen bagud og ind mod kroppen (adduktion) samt indadroterer overarmen.
Pulls the arm backward, adducts it toward the body, and internally rotates it.

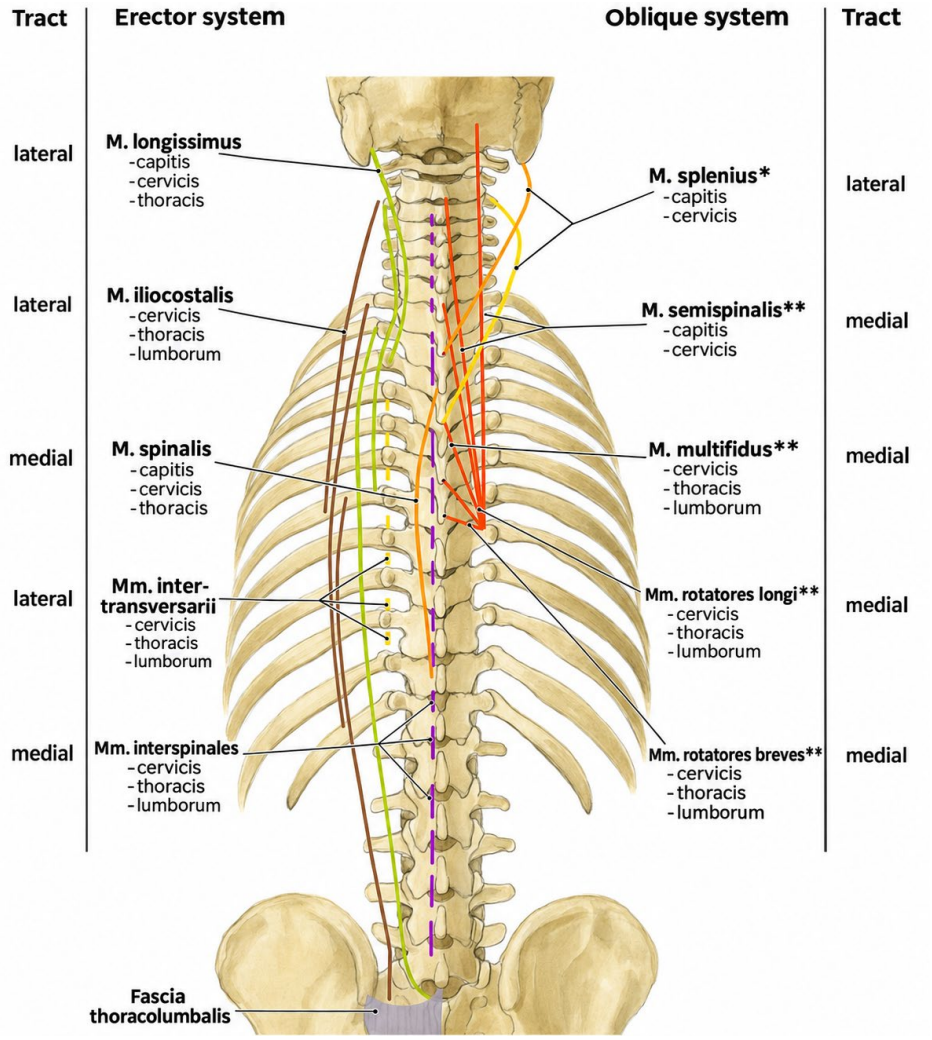
Dybe (autoktone) rygmuskler
Deep (Intrinsic) Muscles of the Back

Laterale muskelgruppe
Lateral Tract

- Tværtapssystem (***mm. intertransversarii***): Segmental stabilisering og lateralfleksion.
Intertransverse system: Segmental stabilization and lateral flexion.

- Sakrospinalt system (***m. iliocostalis, m. longissimus***): Ekstension og lateralfleksion af rygsøjlen.
Sacrospinal system: Spinal extension and lateral flexion.

- Torntap-tværtapssystem (***mm. splenii***): Stabilisering og bevægelse af hoved og nakke.
Spinotransverse system: Stabilization and movement of the head and neck.



Mediale muskelgruppe
Medial Tract

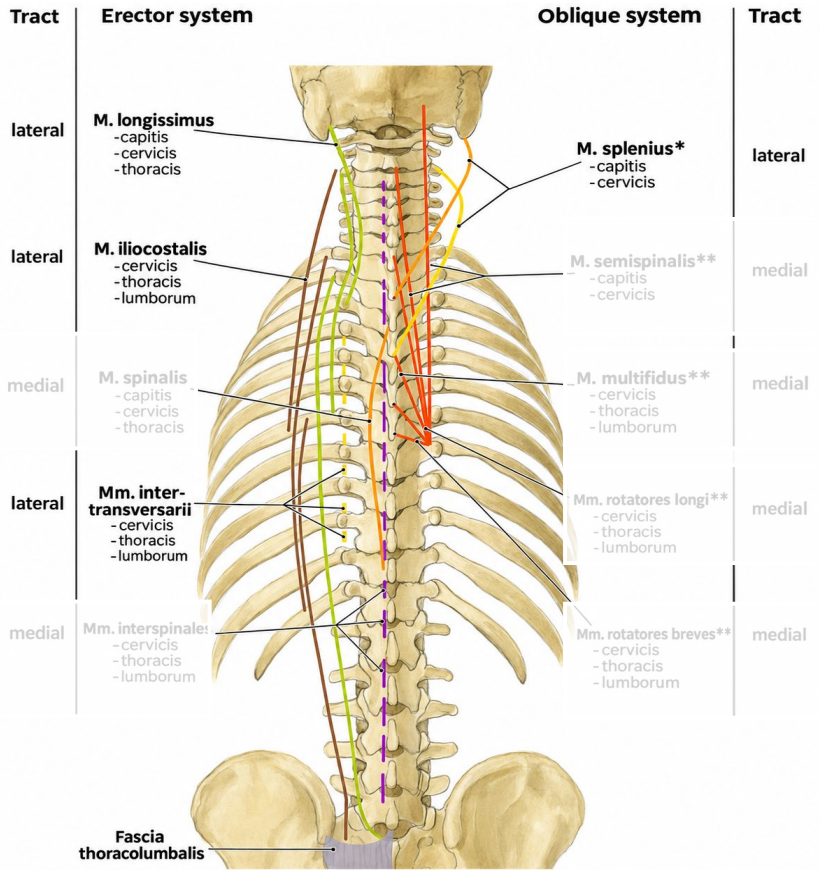
- Torntapssystem (***m. spinalis, mm. interspinales***): Ekstension og segmental stabilisering.
Spinal system: Extension and segmental stabilization.

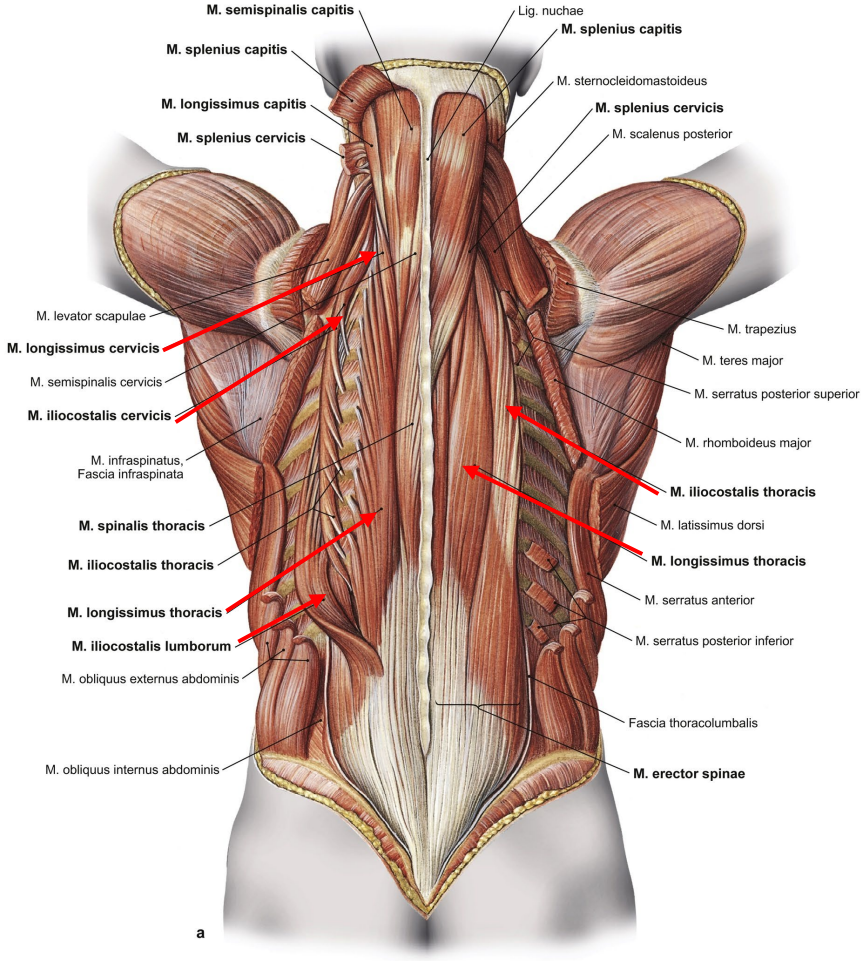
- Tværtap-torntapssystem (***m. semispinalis, m. multifidus, mm. rotatores***): Stabilisering, ekstension og rotation mod modsatte side.
Transversospinal system: Stabilization, extension, and contralateral rotation.

Laterale muskler bidrager især til ekstension og lateralfleksion, mens mediale muskler bidrager til segmental stabilisering og rotation.
Lateral muscles mainly contribute to extension and lateral flexion, while medial muscles contribute to segmental stabilization and rotation.

Laterale muskelgruppe – Anatomi og funktion

Lateral Tract – Anatomy and Function



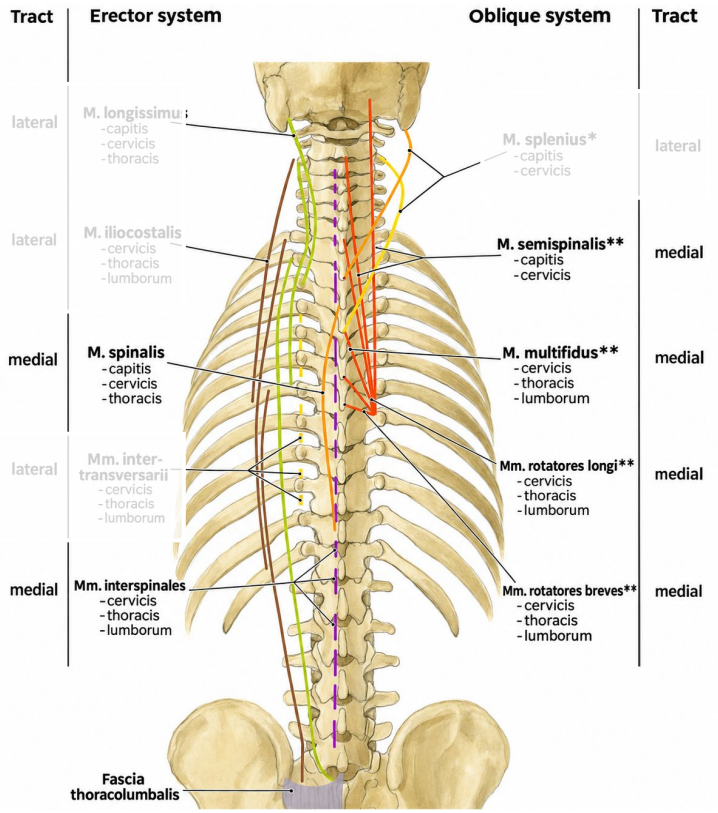


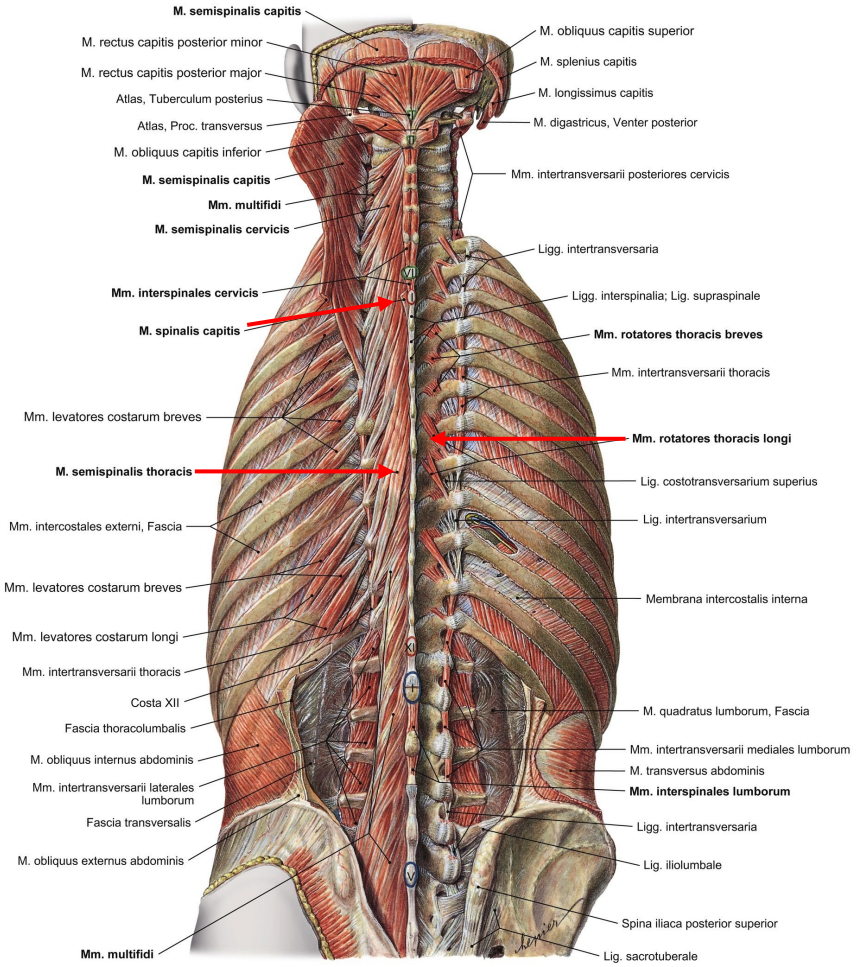
Den laterale muskelgruppe bidrager især til rygsøjlens ekstension, lateralfleksion og stabilitet.

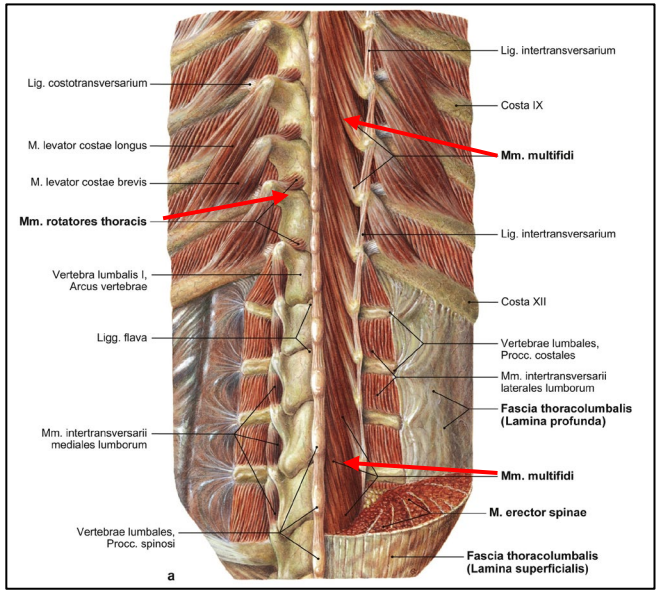
The lateral tract primarily contributes to spinal extension, lateral flexion, and stability.

Mediale muskelgruppe – Anatomi og funktion

Medial Tract – Anatomy and Function





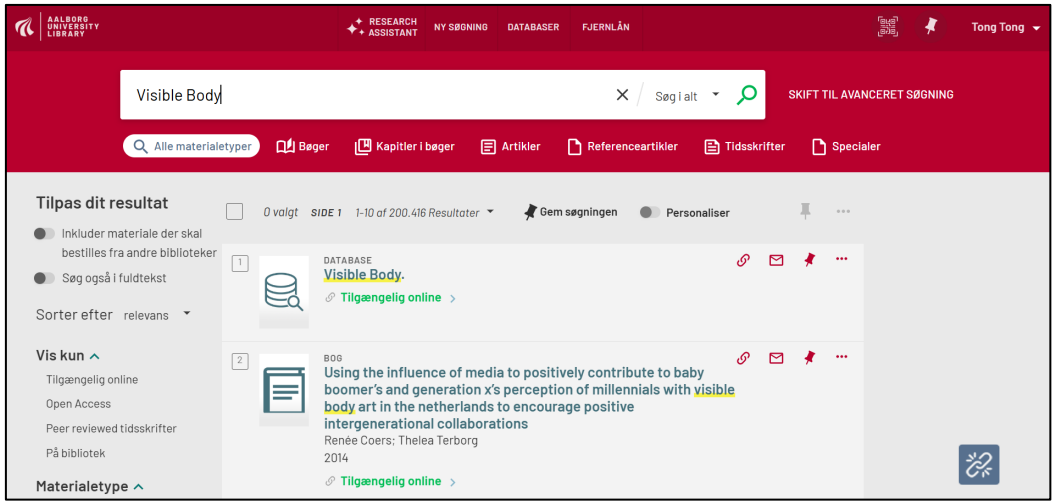


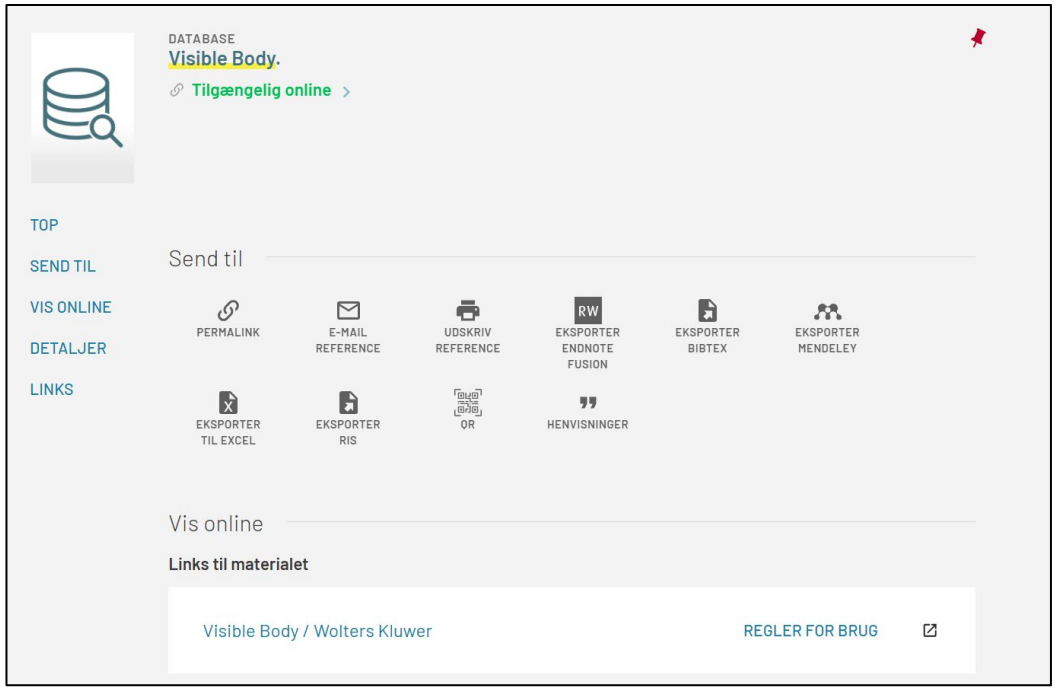
Den mediale muskelgruppe bidrager især til rygsøjlens stabilitet og kontrollerede bevægelser.

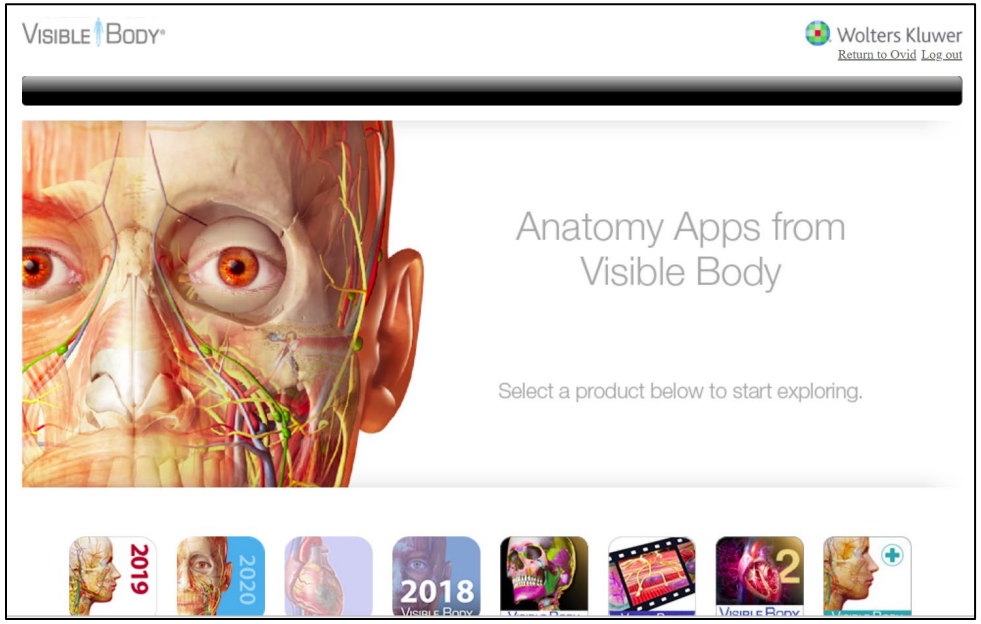
The medial tract contributes primarily to spinal stability and controlled movements.

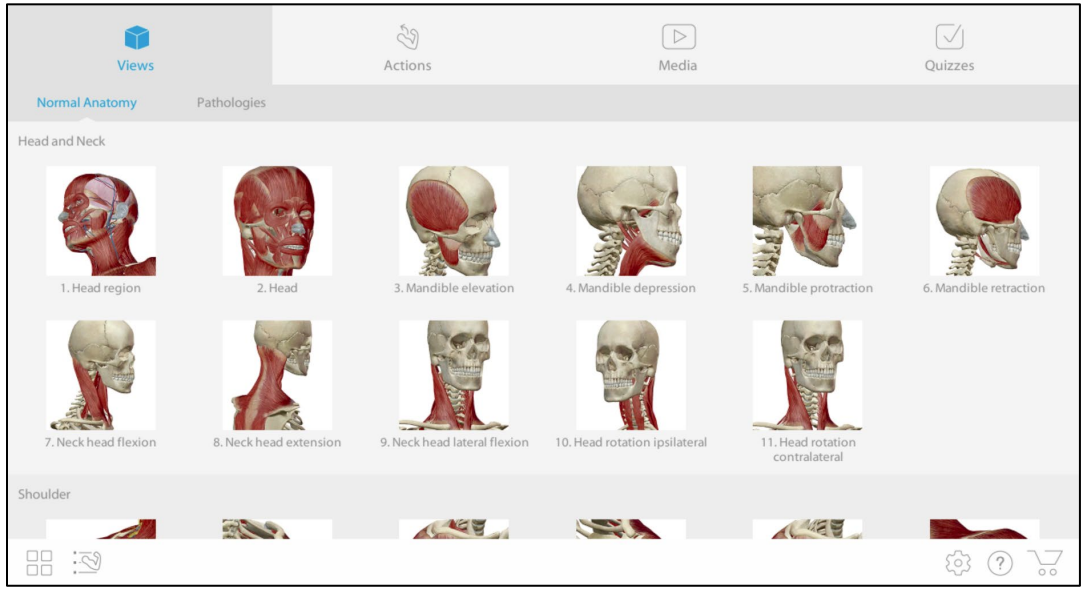
Dybe rygmuskler – Anatomisk inddeling
Intrinsic Back Muscles – Anatomical Classification

Anatomisk gruppe ***Anatomical group***	**Vigtigste muskler** ***Main muscles***	**Hovedfunktion** ***Main function***
Laterale gruppe – Tværtapssystem *Lateral tract – Intertransverse system*	***Mm. intertransversarii***	Segmental stabilisering og lateralfleksion. *Segmental stabilization and lateral flexion.*
Laterale gruppe – Sakrospinalt system *Lateral tract – Sacrospinal system*	***M. iliocostalis, m. longissimus***	Ekstension, lateralfleksion og bidrag til rotation. *Spinal extension, lateral flexion and contribution to rotation.*
Laterale gruppe – Torntap-tværtapssystem *Lateral tract – Spinotransverse system*	***M. splenius capitis, m. splenius cervicis***	Ekstension og bevægelse af hoved og nakke. *Head and neck extension and movement.*
Mediale gruppe – Torntapssystem *Medial tract – Spinal system*	***M. spinalis, mm. interspinales***	Ekstension og segmental stabilisering. *Spinal extension and segmental stabilization.*
Mediale gruppe – Tværtap-torntapssystem *Medial tract – Transversospinal system*	***M. semispinalis, m. multifidus, mm. rotatores***	Stabilisering, ekstension og rotation mod modsatte side. *Stabilization, extension and contralateral rotation.*


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Renée Coers; Thelea Terborg
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Head and Neck
1. Head region
2. Head
3. Mandible elevation
4. Mandible depression
5. Mandible protraction
6. Mandible retraction
7. Neck head flexion
8. Neck head extension
9. Neck head lateral flexion
10. Head rotation ipsilateral
11. Head rotation contralateral
Shoulder

Læringsmål – Systematisk anatomi

Learning Objectives – Systematic Anatomy

- **Beskrive rygsøjlens regioner, hvirvler og fysiologiske krumninger.**

  Describe spinal regions, vertebral morphology, and physiological curves.

- **Forklare facetled, ligamenter og disci samt deres funktion.**

  Explain facet joints, ligaments, discs and their functions.

- **Identificere overfladiske og dybe rygmuskler og deres hovedfunktion.**

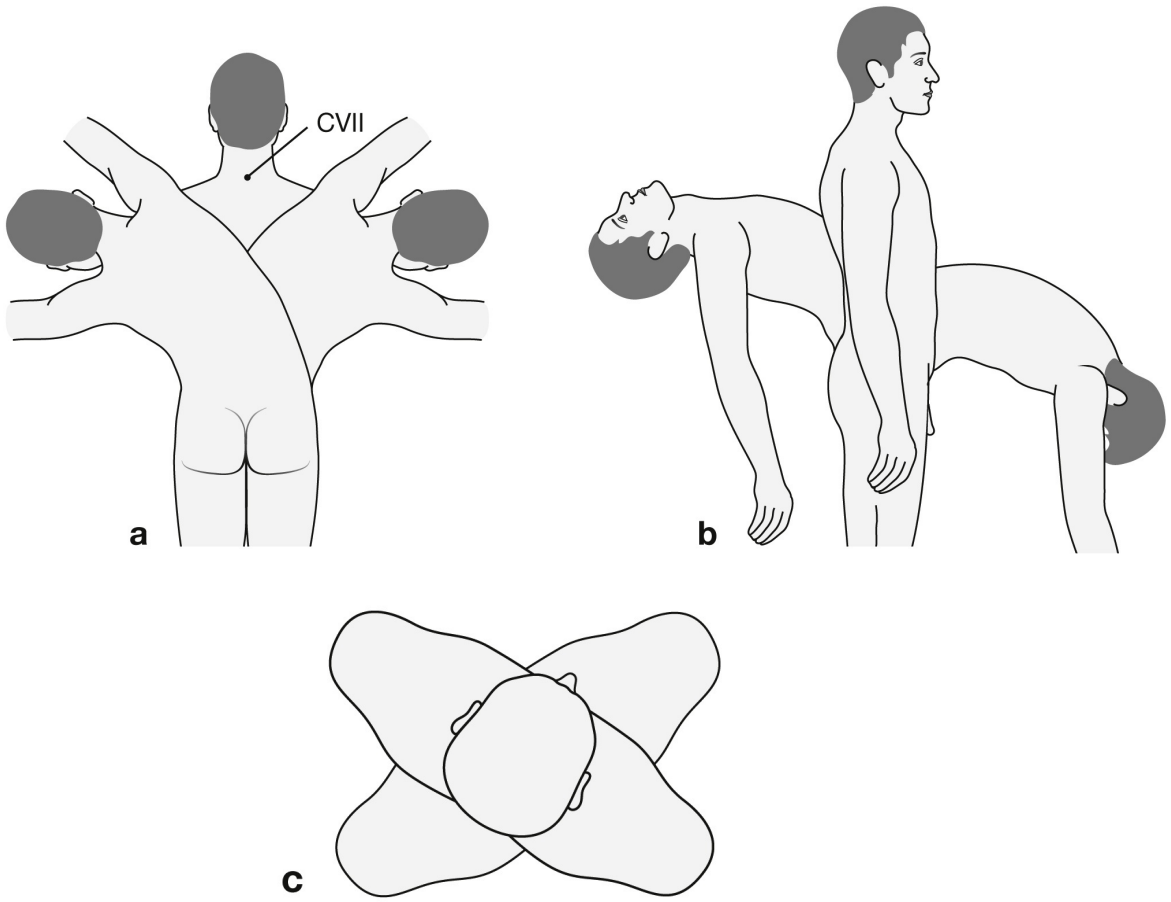
  Identify major extrinsic/intrinsic back muscles and their actions.

- **Forklare fleksion, ekstension, lateralfleksion og rotation.**

  Explain spinal flexion, extension, lateral flexion and rotation.

Rygsøjlens bevægelser – Muskel- og ledkontrol

Spinal Movements – Muscles and Joint Control



- Fleksion (ca. 100°): Bugmuskler bøjer truncus; rygstrækkerne kontrollerer foroverbøjning.
Flexion: Abdominal muscles flex the trunk; spinal extensors control forward bending.

- Ekstension (ca. 50°): M. erector spinae og dybe rygmuskler strækker rygsøjlen.
Extension: Erector spinae and deep back muscles extend the spine.

- Lateralfleksion (ca. 40° til hver side): Unilateral aktivitet i rygstrækkere, bugmuskler og m. quadratus lumborum.
Lateral flexion: Unilateral activity of spinal extensors, abdominal muscles, and quadratus lumborum.

- Rotation (ca. 40° til hver side): Bugmuskler og tværtap-torntapsmuskler; begrænset i lænderyggen.
Rotation: Abdominal and transversospinal muscles; limited in the lumbar spine.

Musklerne skaber og kontrollerer bevægelse, mens facetled, disci og ligamenter bidrager til stabilitet og begrænser bevægeligheden.
Muscles generate and control movement, while facet joints, discs, and ligaments contribute to stability and restrict excessive motion.